

MatterMind: An LLM-Assisted Platform for Crystal Generation, Materials Simulation, and AI-Guided Analysis

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ABSTRACT

The growing complexity of computational materials research calls for integrated platforms that can connect structure generation, first-principles simulation, automated post-processing, visualization, and scientifically grounded interpretation within a unified workflow. Here, we develop MatterMind, a web-based, workflow-aware platform for end-to-end materials research. MatterMind combines a React/FastAPI/Celery/Redis architecture with dedicated execution environments for MatterGen-based crystal generation and VASP-family calculations, and further extends these workflows through structured post-processing, quality control, artifact management, and large-language-model (LLM)-assisted interpretation.

The platform currently supports two major workflow branches: (i) MatterGen-based crystal generation, screening, and descriptor extraction, and (ii) VASP-based HDF5 parsing, VTST/CI-NEB transition-state analysis, Wannier90 localization, and post-Wannier electronic-structure reconstruction. Representative case studies show that MatterMind can transform raw outputs into symmetry-annotated candidate sets, structured DFT summaries, migration-barrier analyses, compact Wannier Hamiltonians, and reusable band/DOS results. These results demonstrate that MatterMind is not merely a job-submission interface, but a unified computational-analysis environment that reduces manual workflow fragmentation and enables structured, visual, and AI-ready materials research.

KEYWORDS AI for Science; Large Language Models; Computational Materials Science; Workflow Orchestration

1 Introduction

Computational materials science is increasingly characterized by multi-stage pipelines rather than isolated simulations: candidate structures may be generated by generative models, refined by density-functional theory (DFT), screened by property workflows, and subsequently analyzed by specialized post-processing (e.g., transport models, defect migration barriers, or Wannier-based tight-binding reductions). Recent diffusion-based generative models such as MatterGen have accelerated the upstream discovery step by directly generating crystal structures under stability and design objectives, strengthening the need for integrated downstream automation that can validate and interpret these candidates with first-principles tools¹.

At the same time, the community has converged on the view that automation must be paired with provenance and structured metadata to ensure that results are traceable, comparable, and reusable across projects and institutions. Database-backed workflow infrastructures explicitly target reproducibility by tracking data provenance as directed graphs and storing inputs/outputs in curated repositories². More broadly, "FAIR for workflows" and packaging standards for workflow-run provenance emphasize that workflows themselves are valuable digital objects whose execution context, parameters, and artifacts must be recorded to enable reuse and quality assessment³.

Despite this progress, day-to-day materials research still often requires manual stitching: converting outputs between tools, re-parsing large text/XML logs, and hand-assembling figures and quality-control (QC) narratives. As the field adopts more diverse pipeline endpoints (e.g., NEB barriers, Wannier Hamiltonians, or ML-generated candidates), the scientific bottleneck moves from "running a code" to "running, validating, and interpreting a workflow end-to-end." This changing bottleneck motivates platforms that are not only workflow executors, but workflow-aware analysis environments.

A large ecosystem of workflow tools has emerged, with different emphases on provenance, scalability, and usability. AiiDA established a widely adopted paradigm of automation plus database-stored provenance graphs, formalized in its original and subsequent major releases². Its process/workflow engine has been described as an event-based infrastructure for robust and modular workflows⁴. Web-accessible ecosystems such as AiiDALab further address usability by packaging workflows and environments as browser-based "apps"⁵

In parallel, the Materials Project ecosystem has developed Python-native workflow stacks centered on FireWorks, jobflow, and atomate2, emphasizing composable Python workflows, structured task documents, and practical execution on diverse resources⁶. Atomate2 explicitly documents schema-driven "Task Documents" for organizing inputs/outputs and enabling downstream processing through related data tooling (e.g., emmet or magma)⁷.

Other frameworks address workflow automation in more lightweight or domain-specific ways. pyiron provides a workflow-oriented integrated development environment with job management and hierarchical data management⁸. The Atomic Simulation Environment offers programmatic building blocks (calculators, structure manipulation, analysis) and includes NEB algorithms and a compact database module (ase.db) with a web interface, supporting smaller-scale automation and post-processing patterns⁹. The diversity of these systems, however, has also made interoperability a first-order concern: a recent proposal for a Python Workflow Definition (PWD) explicitly argues that differing workflow formats hinder interoperability and reproducibility across engines, motivating exchange formats spanning AiiDA, jobflow, and pyiron¹⁰.

Within VASP-centric workflows, parsing reliability and performance are recurring issues because traditional outputs (OUTCAR, vasprun.xml) are large and sensitive to format variations. The newer vaspout.h5 output provides a hierarchical HDF5 representation of inputs and results, but it is only generated when VASP is compiled with HDF5 support, and official guidance recommends dedicated tooling for analysis¹¹. The official Python interface py4vasp reads the HDF5 outputs directly and emphasizes avoiding parsing issues associated with XML/OUTCAR-based pipelines¹². This transition creates both an opportunity (cleaner, structured outputs) and a tooling gap, because many existing workflow stacks remain primarily XML/text oriented.

Transition-state calculations (e.g., diffusion barriers) are essential in catalysis, battery materials, and defect physics and are commonly approached with the Nudged Elastic Band method; the original NEB formulation and the Climbing Image refinement remain standard references. The widely used VTST tools package provides scripts and utilities for saddle-point searches and NEB workflows alongside VASP¹³. Recent workflow frameworks increasingly acknowledge NEB as a core automation target: atomate2 reports that NEB and ApproxNEB workflows for VASP are being added⁷. Meanwhile, AiiDA's VASP plugin includes NEB calculation/workchain components, illustrating how NEB can be embedded in provenance-tracked pipelines

Maximally localized Wannier functions (MLWFs) provide compact localized representations for interpolation and model building, with seminal formalism for localization and disentanglement and an established community codebase¹⁴. However, robust Wannierization remains difficult to automate in general because it often requires user choices for projections and energy windows. Automated protocols have been proposed—e.g., high-throughput Wannierization workflows based on SCDM and curated heuristics—explicitly targeting the parameter sensitivity that hinders large-scale deployment¹⁵. On the software side, Wannier90's documentation defines post-processing modules (postw90) for DOS and k-path analysis and provides

parameterized workflows for band interpolation and DOS generation. For VASP users, the VASP – Wannier90 interface is controlled by tags such as LWANNIER90, which writes the required Wannier input and matrix files, and related tags enabling library-mode execution. AiiDA further supports Wannier workflows through dedicated plugins and workflow packages.

Large language models (LLMs) have recently been studied as assistants for hypothesis generation, tool-augmented experimentation, and scientific reasoning, with both promises and well-documented reliability concerns. Tool-augmented agentic systems illustrate that LLMs can orchestrate complex actions when coupled to search and execution tools¹⁶. At the same time, empirical studies show that LLM-generated scientific summaries can overgeneralize conclusions, motivating caution and grounding mechanisms¹⁷. Surveys and reviews now discuss how LLMs may reshape scientific workflows and how evaluation and traceability should be handled¹⁸. These findings encourage designs where language models operate on structured, provenance-linked outputs rather than free-form logs, and where they expose sources and QC context rather than generating ungrounded narratives.

The literature above demonstrates rapid progress, yet several concrete gaps remain for end-to-end materials workflows:

Limited "workflow-to-analysis" continuity: Many workflow systems prioritize execution and data storage, but users still frequently perform downstream post-processing (plotting, extracting descriptors, summarizing QC) in ad hoc scripts or notebooks. Even where task schemas exist, aligning them across heterogeneous modules remains nontrivial¹⁹.

Uneven support for emerging structured outputs (VASP HDF5). The transition to vaspout.h5 and dedicated extraction tools (py4vasp) improves robustness and structure, but it also creates a tooling discontinuity for pipelines that still assume XML/text parsing¹¹.

Automation gaps in NEB and Wannier pipelines: NEB and Wannier steps are more workflow-sensitive than single-point DFT: they involve multiple dependent calculations, specialized file organization, and additional QC logic (image ordering, convergence consistency, disentanglement stability). Although platforms increasingly implement these workflows, end-to-end integration with artifact bundling and standardized analysis outputs is still inconsistent⁷.

Artifact management and visualization as first-class outputs: Some systems provide web interfaces or database viewers, but generated plots, derived metrics, and publication-ready figures are not uniformly treated as standard workflow artifacts, especially across heterogeneous modules⁵.

Lack of grounded, workflow-aware AI layers: LLMs can assist interpretation, but their tendency to overgeneralize and hallucinate makes it risky to use them as ungrounded narrators of scientific runs. The open research direction is to integrate them with structured provenance, QC, and retrieval-based grounding rather than free-form summarization¹⁷.

These gaps collectively motivate a platform that (i) chains heterogeneous modules, (ii) produces structured metrics and artifacts by default, (iii) exposes workflow-aware QC, and (iv) supports AI-assisted interpretation only when grounded in the workflow's structured outputs.

This manuscript introduces MatterMind, a workflow-aware platform designed to bridge the gaps above by integrating (1) generative materials candidates, (2) first-principles simulations, (3) advanced post-processing pipelines, and (4) grounded interpretation within a single web-accessible environment. The prototype implementation of MatterMind has been released as an open-source project on GitHub to improve transparency and reproducibility.

Contribution 1: Heterogeneous workflow chaining with explicit artifact contracts. MatterMind treats each module—materials generation, DFT, NEB, and Wannierization—not as isolated jobs but as linked stages with

typed inputs/outputs and explicit reuse of upstream artifacts (structures, wavefunctions, checkpoints, intermediate images). This aligns with recent arguments that workflow interoperability is hindered by divergent workflow formats and that reusable workflow objects should be shareable independently of the execution system.

Contribution 2: HDF5-first VASP post-processing. MatterMind natively supports VASP compiled with HDF5 to produce vaspout.h5, and it leverages official extraction tooling to generate structured results and plots, consistent with VASP's recommendation to analyze vaspout.h5 via py4vasp. This design is aimed at improving robustness relative to brittle XML/text parsing.

Contribution 3: Turnkey VTST/CI-NEB automation with workflow-aware QC. MatterMind integrates NEB pipelines using established NEB/CI-NEB methodology and VTST tooling, packaging the multi-image directory structure, barrier extraction, and QC signals (e.g., convergence and image integrity) as first-class outputs rather than by-products.

Contribution 4: Wannier90/postw90 orchestration plus compact-Hamiltonian analysis. MatterMind runs VASP→Wannier90 interface steps (e.g., LWANNIER90 and related controls), performs Wannier localization, exports real-space Hamiltonians, and executes post-Wannier band/DOS reconstruction using postw90 modules, converting outputs into structured summaries (spreads, centers, hopping statistics) and publication-ready figures.

Contribution 5: Visualization and LLM-assisted interpretation grounded in structured outputs. MatterMind automatically renders consistent plots (structures, band/DOS, NEB barriers, Wannier centers/hoppings) and exposes structured metrics for programmatic consumption (e.g., JSON summaries). A built-in LLM layer generates human-readable narratives that can be audited against those metrics, motivated by evidence that LLM summaries can overgeneralize scientific results when ungrounded.

Following this Introduction, the manuscript describes (i) platform architecture and workflow contracts, (ii) module implementations for generation, VASP-HDF5, VTST/NEB, and Wannier/postw90, and (iii) validation case studies demonstrating reproducible execution, artifact reuse, QC interpretation, and analysis-ready outputs across these heterogeneous workflows.

Table 1. Comparison of MatterMind with representative workflow ecosystems used in computational materials science.

The table emphasizes capabilities required for end-to-end, workflow-aware analysis: VASP HDF5 parsing, NEB and Wannier automation, artifact management, structured QC, visualization, and grounded LLM assistance. "Partial" denotes functionality that exists but is not a documented default across typical installations or requires additional user scripting.

Platform	HDF5 parsing (VASP vaspout.h5)	NEB automation	Wannier integration	Artifact management	QC metrics / error handling	Visualization	LLM assistance	License
MatterMind (this work)	Yes (HDF5-first via py4vasp)	Yes (VTST/CI- NEB pipeline)	Yes (VASP– Wannier90 + postw90)	Yes (workflow artifact bundles + summaries)	Yes (workflow- aware QC summaries)	Yes (auto plots)	Yes (grounded narratives)	Unspecified (this manuscript)
AiiDA	Partial/unspecified for VASP HDF5 (plugin-dependent); strong provenance store	Yes (NEB in AiiDA-VASP workchains)	Yes (aiida- wannier90 + related workflows)	Yes (graph provenance + repository)	Yes (workflow logic + plugin checks)	Partial (via AiiDALab / external notebooks)	No	MIT
atomate2 / jobflow	Primarily XML/text parsing via task documents (vasprun/OUTCAR), not HDF5-first	In progress / partial (NEB & ApproxNEB being added)	No standard Wannier90 pipeline documented	Yes (schema "Task Documents", DB- ready)	Yes (custodian error handlers)	Partial (user-side plotting)	No	Modified BSD (atomate2)
pyiron	Unspecified (code- and user-dependent)	Partial (supported in some workflows; user- defined)	Unspecified / nonstandard	Yes (hierarchical data management)	Partial (user- defined)	Partial	No	BSD-3-Clause
ASE	Not applicable as a DFT workflow engine; calculator- specific	Yes (NEB implementation)	No	Partial (ase.db for structures + metadata)	Limited	Partial	No	LGPL

2 Methods

2.1 System Architecture

MatterMind was implemented as a layered web-based orchestration platform for end-to-end materials research workflows, including crystal generation, first-principles simulation, automated post-processing, and LLM-assisted interpretation. The architecture was designed to reconcile three practical requirements: a unified user interaction model, workload-specific runtime isolation, and structured downstream analysis. To this end, the system is organized into four functional layers: a presentation layer, a control and orchestration layer, an execution layer, and an analysis layer, as illustrated in Fig. 1. The current research prototype has been released on GitHub.

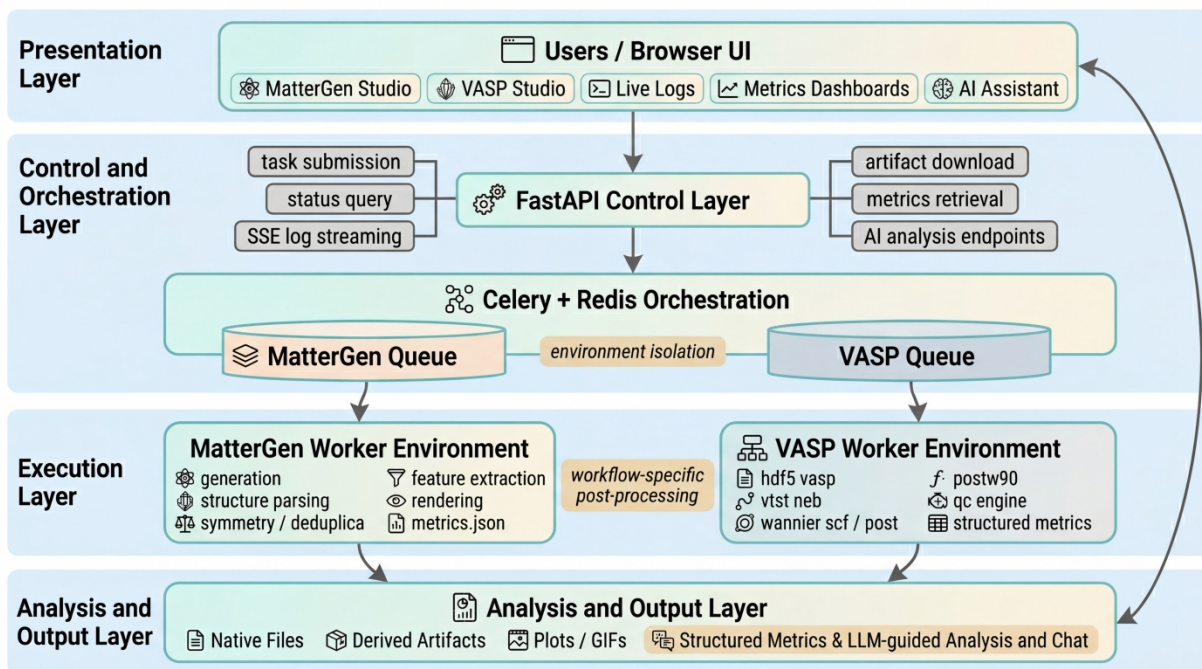


Figure 1. System architecture of the MatterMind platform.

MatterMind is organized as a layered web-based orchestration system for end-to-end materials research workflows. The presentation layer provides a unified browser interface for task submission, live log monitoring, artifact inspection, metrics visualization, and AI-assisted interaction. A FastAPI-based control layer manages task submission, status querying, artifact retrieval, structured metrics access, and server-sent event (SSE) log streaming. At the orchestration layer, Celery and Redis coordinate asynchronous job scheduling through two isolated queues for MatterGen and VASP-family workloads, thereby avoiding dependency conflicts while preserving a unified API. The execution and analysis layers couple workflow-specific computation with automated post-processing, quality control, derived artifact generation, structured metrics extraction, and LLM-guided interpretation.

At the presentation layer, MatterMind provides a browser-based interface built with React and Vite, through which users can submit computational tasks, monitor job status, inspect live logs, browse derived artifacts, and interact with AI-generated reports. The frontend communicates with a FastAPI backend that acts as the central control plane and exposes endpoints for task submission, status querying, artifact retrieval, structured metrics access, and AI analysis services. Live execution feedback is streamed to the browser through server-sent events (SSE), enabling lightweight one-way delivery of runtime updates from the server to the client over HTTP.

The control and orchestration layer is implemented using FastAPI together with Celery and Redis to support asynchronous job dispatch, task-state management, and fault-tolerant workflow scheduling. In this design,

Celery workers consume tasks from message queues coordinated through Redis, which allows long-running scientific workloads to be decoupled from interactive web requests. A key architectural choice in MatterMind is queue-level environment isolation: MatterGen jobs are routed to a dedicated worker environment, whereas VASP-family workflows are dispatched to a separate worker environment. This separation prevents dependency conflicts between machine-learning and electronic-structure software stacks while preserving a unified backend API. In addition, each submitted job is assigned an independent working directory that stores raw inputs, runtime logs, native outputs, derived artifacts, and structured metadata in a traceable and reproducible manner.

At the execution layer, MatterMind supports heterogeneous scientific workloads spanning MatterGen-based crystal generation and VASP-family simulations, including standard HDF5-enabled VASP calculations, VTST/NEB workflows, Wannier workflows, and postw90-derived property calculations. MatterGen provides a generative framework for inorganic crystal design under composition or property-related conditions, whereas VASP HDF5 outputs can be programmatically accessed through py4vasp-compatible parsing routines. These workflow-native outputs are not directly exposed as isolated files alone; instead, they are passed to workflow-specific post-processing modules for parsing, validation, and transformation.

The analysis layer couples automated scientific post-processing with structured interpretation. Native outputs are converted into task-specific derived artifacts, including quality-control indicators, visualization products, and structured metrics files. These metrics serve as a common intermediate representation between computation, dashboard rendering, and LLM-based reasoning, allowing the same execution result to support both human inspection and context-aware AI interpretation. To provide this interpretation capability, MatterMind connects to the Alibaba Cloud Bailian platform and currently supports API-based access to five state-of-the-art general-purpose foundation models: Qwen3.6-Plus²⁰, DeepSeek-V3.2²¹, Kimi-K2.5²², GLM-5²³, and MiniMax-M2.5²⁴. Within the platform, these models are not used as unrestricted chat agents, but are guided through task-specific prompting and structured workflow outputs so that their analysis remains grounded in the computational results. In this way, structured interpretation is embedded into the platform architecture as an integral component of the computational workflow rather than being treated as an external downstream utility.

2.2 MatterGen Module

The MatterGen module in MatterMind was implemented as a web-based generative workflow for crystal structure proposal, post-generation screening, visualization, and LLM-assisted interpretation. Rather than exposing the MatterGen model family through a standalone generation interface alone, the module couples conditional or unconditional crystal generation with automated structure normalization, crystallographic annotation, redundancy reduction, descriptor extraction, structured reporting, and downstream candidate triage. In this way, the module functions as a generative front end for the broader materials-discovery workflow of MatterMind, as illustrated in Fig. 2. MatterGen itself is an official generative framework for inorganic materials design across the periodic table and supports steering generation toward a range of property constraints.

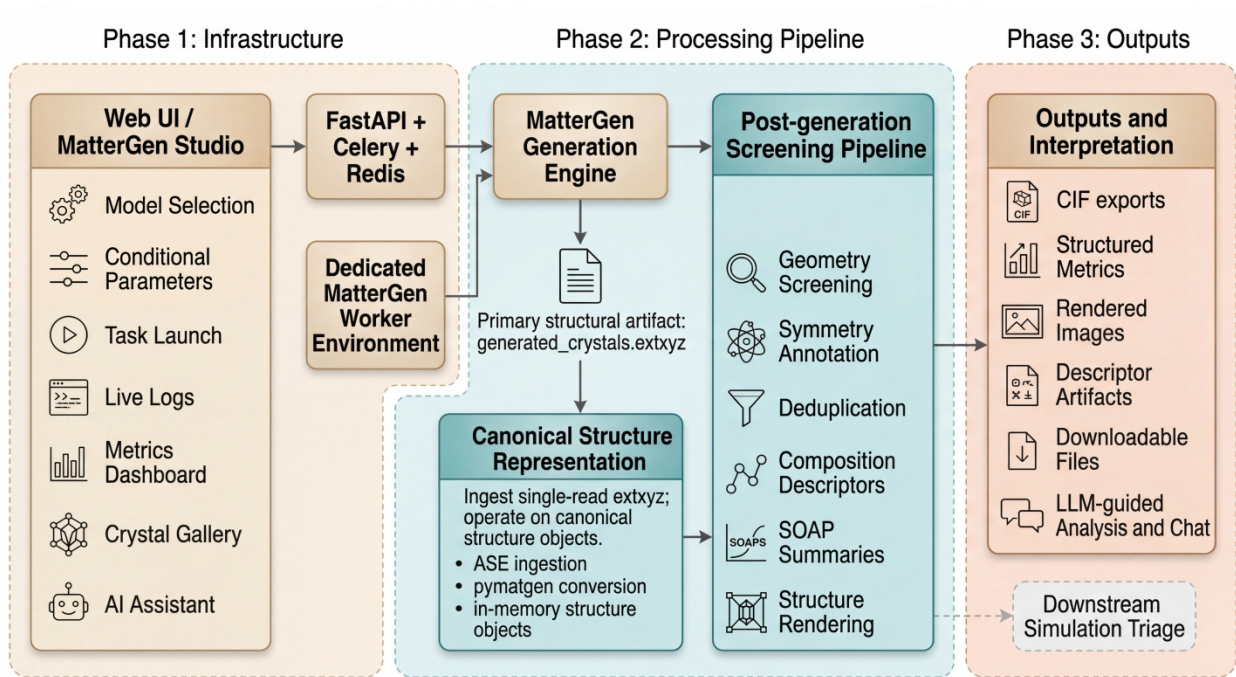


Figure 2. Workflow architecture of the MatterGen module in MatterMind.

The MatterGen module is organized as a workflow-oriented generative subsystem that integrates model selection, conditional crystal generation, isolated execution, post-generation structural normalization, geometric and crystallographic screening, deduplication, descriptor extraction, visualization, structured metrics generation, and LLM-assisted interpretation within a unified web interface. Raw generation outputs are ingested from the primary extxyz artifact and converted into canonical in-memory structure objects for downstream analysis. The resulting task-specific metrics and derived artifacts support candidate triage, dashboard rendering, and AI-guided follow-up recommendations.

At the interaction and orchestration levels, users access the module through the MatterMind web interface for model selection, parameter configuration, task submission, log inspection, artifact download, and AI-guided analysis. MatterGen tasks are routed through the common FastAPI-Celery-Redis backend described in Section 2.1, but are dispatched to a dedicated queue and executed in an isolated MatterGen-specific worker environment. This separation is necessary because the generative workflow depends on a machine-learning software stack that is distinct from the VASP-oriented runtime used for first-principles calculations. Each generation task is executed in an independent working directory so that raw outputs, derived artifacts, runtime logs, and structured metadata remain traceable and reproducible throughout the post-processing pipeline.

The execution core of the module is the MatterGen generation engine, which is invoked programmatically within a per-job working directory. MatterMind supports both unconditional and conditional generation modes. In the unconditional setting, base checkpoints such as `mattergen_base` and `mp_20_base` are used to sample candidate crystal structures without explicit target properties. In the conditional setting, the interface exposes chemistry-, symmetry-, and property-constrained generation, including chemical-system, space-group, magnetic-density, band-gap, bulk-modulus, and multi-property conditioning modes supported by the MatterGen model family. User-defined targets are translated into model-compatible generation arguments, including property-conditioning fields and guidance-strength parameters, so that a broad range of generative settings can be accessed through a uniform submission interface. MatterGen's official command-line workflow documents both the pretrained model families and the use of `properties_to_condition_on` together with `diffusion_guidance_factor` for conditional generation.

A central design choice in the MatterGen module is that the generated extxyz output is treated as the primary structural artifact for post-generation analysis. According to the official MatterGen workflow,

generated_crystals.extxyz stores the generated structures as frames, alongside optional trajectory archives and per-structure CIF exports. In MatterMind, this extxyz output is ingested once using ASE and converted into in-memory structure representations, after which downstream analyses operate on canonical pymatgen Structure objects rather than repeatedly re-reading intermediate files. This design is consistent with ASE's support for Extended XYZ as a frame-based format carrying lattice and per-frame metadata, and with pymatgen's AseAtomsAdaptor, which provides direct conversion between ASE Atoms objects and pymatgen structure objects.

Following structural ingestion, the module performs automatic geometric and crystallographic screening. For each generated candidate, MatterMind derives composition, reduced formula, lattice descriptors, cell volume, and periodic-boundary metadata, and performs rapid plausibility checks based on minimum interatomic distances, nearest-pair identification, and close-contact counting under user-defined thresholds. In parallel, symmetry metadata are extracted with spglib, including the international space-group number, international short symbol, Hall symbol, and equivalent-atom mappings. To reduce redundancy among sampled candidates, structurally similar outputs are grouped with pymatgen's StructureMatcher, and the resulting group assignments are written into the task-level structured summary as an automated diversity filter prior to more expensive downstream evaluation. Spglib explicitly exposes symmetry dataset fields such as number, international, hall, and equivalent_atoms, while pymatgen documents StructureMatcher and its group_structures() functionality within the structure-matching module.

The module further enriches each candidate with lightweight descriptor-based characterization. Composition-level features are generated with matminer using selected Magpie-style summary statistics, while local structural environments are represented through periodic SOAP descriptors computed with DScribe. Large descriptor arrays are preserved as downloadable artifacts, whereas compact statistical summaries are retained in the central metrics record to keep the main structured output concise. In addition, generated candidates are exported into per-structure crystallographic files and rendered through an automated visualization pipeline for direct inspection in the web dashboard. Matminer documents Magpie-based composition summaries, and DScribe defines SOAP as a descriptor based on a local expansion of Gaussian-smeared atomic density with orthonormal radial and angular basis functions.

The final output of the MatterGen post-processing workflow is a task-specific structured metrics.json file that aggregates job metadata, provenance information, geometry summaries, symmetry annotations, redundancy-reduction results, descriptor coverage, and warning flags. This file is designed as a machine-readable scientific screening summary rather than as an unstructured log dump. In the user interface, the same structured record is rendered into metrics panels, per-structure tables, and a crystal gallery of derived visualizations, while native and derived artifacts remain available for download. By grounding the LLM layer in this structured evidence rather than in raw generation logs, the MatterGen module enables evidence-based candidate triage and workflow-aware follow-up suggestions for downstream simulation and analysis.

2.3 VASP Module

The VASP module in MatterMind was implemented as a workflow-oriented interface for first-principles calculations, transition-state analysis, Wannierization, and post-Wannier property evaluation. Instead of exposing these tasks as isolated file-based executions, the module integrates computation, automated post-processing, quality control, structured metrics generation, visualization, and LLM-assisted interpretation within a unified web environment. All VASP-family workflows share a common user interface and backend control logic, while their execution is confined to a dedicated VASP worker environment that is separated from the MatterGen runtime to avoid dependency conflicts between machine-learning and electronic-structure toolchains. The overall design of this module is illustrated in Fig. 3.

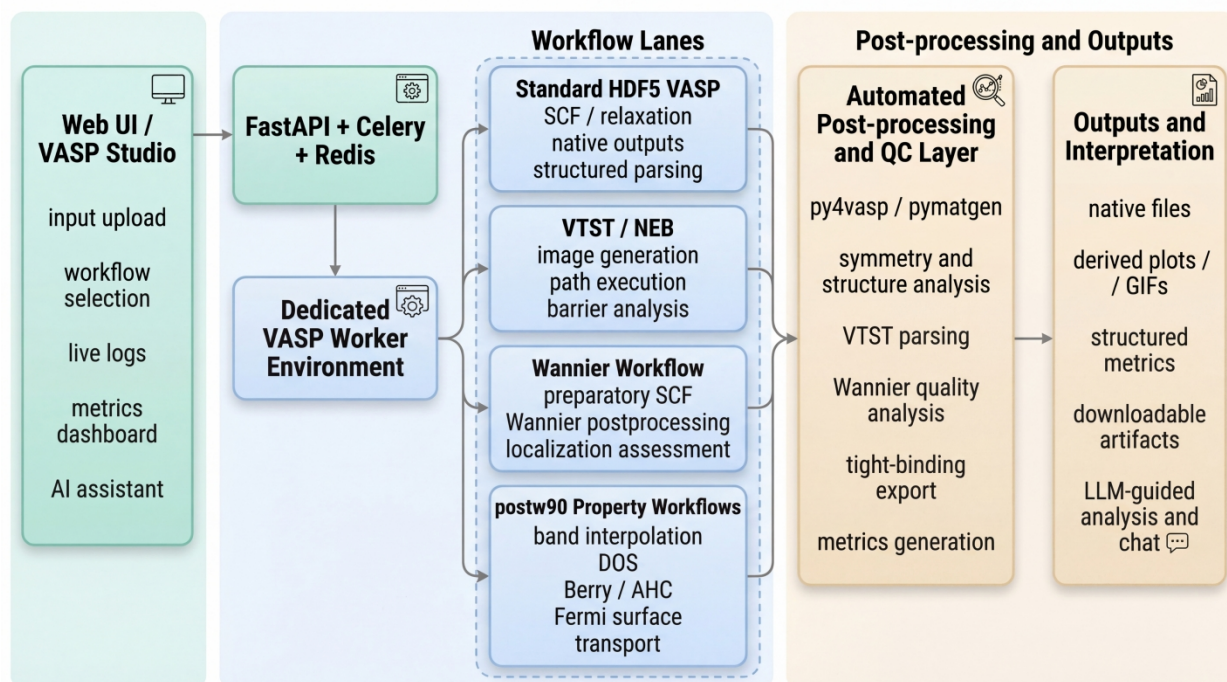


Figure 3. Workflow architecture of the VASP module in MatterMind.

The VASP module is organized as a workflow-oriented subsystem that integrates task submission, asynchronous orchestration, isolated execution, automated post-processing, quality control, structured metrics generation, and LLM-assisted interpretation within a unified web interface. Standard HDF5-enabled VASP calculations, VTST/NEB workflows, Wannier workflows, and post-Wannier property modules are dispatched through a shared control layer to a dedicated VASP worker environment. Native outputs are subsequently transformed into derived visualizations, workflow-specific quality indicators, and structured metrics files that support both interactive dashboard rendering and AI-guided analysis.

At the control and orchestration level, the VASP module is managed by the FastAPI-Celery-Redis backend described in Section 2.1. User-submitted tasks are routed to a dedicated VASP queue and executed in independent job directories under a persistent task root, thereby preserving the full computational context of each run, including uploaded inputs, runtime logs, native outputs, derived artifacts, and structured metadata. The web interface supports workflow-specific input submission and provides real-time log streaming through server-sent events, enabling users to monitor long-running calculations without direct shell access.

The module currently supports four workflow families, each associated with a dedicated execution and post-processing pipeline. The first is a standard HDF5-enabled VASP workflow for self-consistent field, structural relaxation, and related single-directory calculations. In this mode, workflow-native outputs are parsed into a unified structured summary that includes structural, energetic, force, stress, magnetic, and electronic information, together with crystallographic normalization and workflow-specific quality flags. This design is consistent with the HDF5-based post-processing model recommended in the VASP ecosystem, where `vaspout.h5` is intended for structured analysis and `py4vasp` provides direct access to the HDF5 output.

The second workflow family targets transition-state and migration-path calculations based on the VTST toolkit. In this case, MatterMind automates the preparation and execution of nudged elastic band calculations and transforms the resulting path information into structured summaries of barrier profiles, image-wise energy and force trends, transition-state candidates, and path-level quality diagnostics, together with derived visualizations for result inspection. The third workflow family is devoted to Wannierization and is separated into a preparatory SCF stage and a subsequent Wannier post-processing stage. This workflow generates the interface files required by `wannier90.x` and performs a dedicated quality-assessment pass that summarizes quantities such as the number of Wannier functions, final centers, spreads, spread decomposition terms, convergence

status, warnings, and derived localization-quality labels. These design choices align with the intended roles of VTST for saddle-point and minimum-energy-path calculations, and of Wannier90 for maximally localized Wannier functions and related tight-binding-style representations.

The fourth workflow family extends completed Wannier calculations into post-Wannier property evaluation. From a successful Wannier task, users can launch property-specific downstream analyses without re-uploading the full input set. The system programmatically updates the shared Wannier input context and invokes the appropriate downstream engine for modules such as band interpolation, density of states, Berry-curvature-related quantities, Fermi-surface analysis, and BoltzWann transport. Each subworkflow produces its own structured metrics file and derived plots, thereby preserving modularity at the execution level while maintaining a common interaction model at the interface level. These capabilities are consistent with the documented Wannier90/postw90 feature set for band structures, density of states, Fermi surfaces, and advanced electronic-property calculations.

A central methodological feature of the VASP module is that quality control is embedded into execution rather than applied retrospectively. For standard calculations, MatterMind evaluates clean termination, electronic and ionic convergence, force and stress thresholds, output completeness, and common runtime error signatures. For VTST workflows, it additionally checks image-count consistency, endpoint availability, convergence quality across images, and the physical plausibility of the path shape. For Wannier-related workflows, it evaluates localization quality, spread magnitude, orbital-center offsets, and the compactness of the derived tight-binding representation. These quality indicators are written directly into the structured metrics and exposed in the interface as readable status summaries.

Finally, the VASP module interfaces directly with the analysis layer of MatterMind through workflow-specific structured metrics and derived artifacts. As a result, the LLM component operates on curated computational evidence rather than on unfiltered raw output text, enabling task-aware interpretation for standard DFT, NEB, Wannier, and post-Wannier property workflows. Accordingly, the VASP module serves as an integrated computational-analysis subsystem in which execution, validation, summarization, visualization, and AI-assisted reasoning are combined within a single reproducible environment.

3 Results and Discussion

3.1 Mattergen Calculation

3.1.1 Case 1: Unconditional crystal generation with `mattergen_base`

To establish a baseline for the MatterGen workflow implemented in MatterMind, we first performed an unconditional crystal-generation experiment using the `mattergen_base` model with `batch_size = 4` and `num_batches = 4`, yielding 16 candidate structures in total. The purpose of this case was not to claim materials discovery at this stage, but to evaluate whether the MatterMind MatterGen module could reliably execute an end-to-end unconditional generation workflow and convert raw outputs into structured, screenable candidates for downstream prioritization.

The run completed successfully and all 16 generated structures propagated through the full post-processing chain. In each case, the generated frames were parsed from `generated_crystals.extxyz`, converted into canonical pymatgen Structure objects, annotated with crystallographic and geometric metadata, exported as CIF files, rendered as structure images, and enriched with Magpie-derived composition features and SOAP-based structural summaries. No job-level warnings were recorded, and full downstream coverage was achieved for all 16 structures, including CIF export, PNG rendering, SOAP summaries, and Magpie feature generation. This result indicates that the MatterMind pipeline can transform raw generative outputs into consistent, machine-readable scientific summaries without manual intervention.

From the standpoint of geometric plausibility, the entire batch passed the initial screening criteria implemented in the platform. No generated structure exhibited close contacts below the preset threshold of 1.5 Å, and no per-structure geometry warnings were triggered. The smallest minimum interatomic distance in the batch was 1.84 Å, observed for $\text{Rb}_3\text{TmCu}_2\text{F}_{12}$, whereas most candidates exhibited larger values in the approximate range of 2.0-3.3 Å. Although such screening does not establish thermodynamic stability, it does indicate that the generated candidates are not trivially unphysical at the level of short-range geometry and are therefore suitable for first-pass downstream screening.

The unconditional batch also showed no evidence of trivial structural redundancy at this sampling scale. Under the applied StructureMatcher tolerances ($\text{ltol} = 0.2$, $\text{stol} = 0.3$, and $\text{angle_tol} = 5^\circ$), all 16 structures were assigned to distinct deduplication groups. This observation suggests that, within this small batch, the unconditional workflow did not repeatedly sample near-identical motifs and instead retained a useful degree of structural diversity for subsequent triage. Because the present case involves only 16 candidates, this finding should be interpreted as a workflow-scale demonstration of non-redundancy rather than as a population-level characterization of the underlying generative model.

Crystallographically, the batch covered a broad symmetry range. Five structures were assigned to the triclinic space group $P1$ two to $P\bar{1}$ two to Cm , and two to the cubic space group $F\bar{4}3m$ while the remaining candidates were distributed among $C2/c$, $P4mm$, $R3m$, $P4/mmm$, and $Amm2$. Within this small sample, low-symmetry outputs were numerically dominant, but higher-symmetry candidates were also observed. This mixture is desirable for exploratory generation, because it combines structural variety with the presence of more crystallographically regular candidates that may be attractive for subsequent relaxation-based evaluation.

Representative structures from this unconditional run are shown in Fig. 4, and a structured screening summary for eight selected candidates is provided in Table 2. The selected examples were chosen to span both chemically compact and compositionally complex regions of the generated design space. Among them, FeCo was generated in the rhombohedral space group $R3mR3mR3m$ with the smallest volume per atom in the batch ($11.62 \text{ \AA}^3/\text{atom}$), making it the densest candidate observed. ZrTiFeCu and TbCu_4Ru both crystallized in the cubic space group $F\bar{4}3m$ with volumes per atom of 15.30 and $14.67 \text{ \AA}^3/\text{atom}$, respectively. These structures combine relatively high symmetry with compact packing and therefore appear to be pragmatic first-pass targets for more expensive relaxation and energetic evaluation. Notably, FeCo and ZrTiFeCu were also selected as Top-K candidates by the integrated AI analysis function of MatterMind, in which Qwen3.5-plus operated on the task-specific metrics.json generated by the platform rather than on raw logs. This agreement between rule-based screening indicators and metrics-grounded AI prioritization provides an initial example of how structured post-processing and LLM-assisted analysis can converge on similar follow-up recommendations.

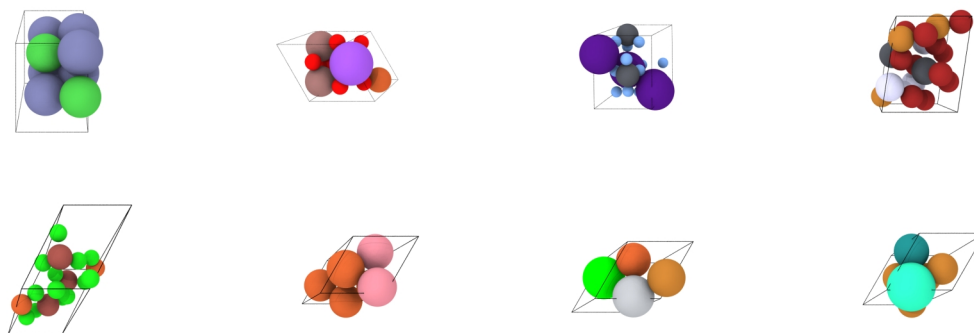


Figure 4. Representative crystal structures generated in the unconditional mattergen_base case and rendered by the MatterMind post-processing pipeline.

Eight representative candidates were selected from the 16 generated structures to illustrate the structural and chemical diversity of the unconditional generation workflow, including compact intermetallic-like candidates (Zn₃Ni, FeCo, ZrTiFeCu, and TbCu₄Ru), oxide/fluoride-containing structures (Na₂In₂FeO₆ and Cs₂PbF₆), and more compositionally complex or spatially open halide-containing candidates (Cu₃Ag₃(PbBr₆)₂ and Tl₃Fe₂Cl₁₃). Together, these examples highlight the ability of the MatterMind MatterGen workflow to transform raw unconditional generative outputs into directly inspectable candidates spanning both compact and exploratory regions of structural space.

Table 2. Screening summary of eight representative structures generated in the unconditional mattergen_base case.

For each selected candidate, the table reports composition, atom count, symmetry assignment, volume per atom, minimum interatomic distance, close-contact statistics, deduplication-group identity, SOAP and Magpie feature coverage, and warning status. The table is intended to summarize how MatterMind converts raw generative outputs into structured evidence for geometric screening, diversity assessment, and downstream prioritization.

#	Formula	Atoms	Symmetry	V/atom	Min dist	Close contacts	Dedup	SOAP	Magpie	Warn
1	Zn ₃ Ni	8	15 C2/c	12.95	2.57 (Zn-Zn)	0	G3 x1	92640	Yes	0
2	Na ₂ In ₂ FeO ₆	11	2 P-1	13.08	2.03 (Fe-O)	0	G12 x1	92640	Yes	0
3	Cs ₂ PbF ₆	18	1 P1	25.66	2.11 (F-Pb)	0	G15 x1	92640	Yes	0
4	Cu ₃ Ag ₃ (PbBr ₆) ₂	20	1 P1	30.54	2.38 (Br-Cu)	0	G0 x1	92640	Yes	0
5	Tl ₃ Fe ₂ Cl ₁₃	18	1 P1	37.06	2.14 (Cl-Fe)	0	G4 x1	92640	Yes	0
6	FeCo	8	160 R3m	11.62	2.45 (Co-Fe)	0	G1 x1	92640	Yes	0
7	ZrTiFeCu	4	216 F-43m	15.3	2.71 (Fe-Zr)	0	G5 x1	92640	Yes	0
8	TbCu ₄ Ru	6	216 F-43m	14.67	2.47 (Cu-Cu)	0	G8 x1	92640	Yes	0

By contrast, several low-symmetry outputs occupied more open and chemically elaborate regions of structural space. For example, Tl₃Fe₂Cl₁₃ exhibited a substantially larger volume per atom (37.06 Å³/atom), while Cu₃Ag₃(PbBr₆)₂ also showed a comparatively expanded structure (30.54 Å³/atom). Such candidates remain useful from a generative-screening perspective because they demonstrate that the unconditional workflow explores not only compact metallic or intermetallic-like structures but also more complex ionic or framework-like motifs. At the same time, their larger apparent openness makes them natural candidates for lower prioritization in the first round of follow-up calculations.

Taken together, this baseline case demonstrates that even the simplest unconditional generation mode already benefits substantially from the MatterMind workflow. The main contribution of the platform in this setting is not merely the execution of MatterGen itself, but the immediate transformation of raw generated structures into a structured and inspectable candidate set with geometric screening results, symmetry assignments, deduplication labels, descriptor coverage, visualization artifacts, and AI-readable summaries. This allows the user to move directly from generative sampling to evidence-based downstream triage. In the present batch, compact and relatively symmetric structures such as FeCo, ZrTiFeCu, and TbCu₄Ru emerge as the most reasonable first targets for subsequent CHGNet/MatGL relaxation or DFT validation, whereas low-symmetry and high-volume candidates can be deprioritized or retained as exploratory outliers.

A limitation of this baseline experiment is that it remains a geometry- and symmetry-level screening case rather than a full stability assessment. No formation energies, relaxed energies, band gaps, or magnetic properties were available in the present structured metrics, and therefore no strong claim can be made regarding thermodynamic realizability. Accordingly, Case 1 should be understood as a baseline workflow demonstration showing that MatterMind can produce a non-redundant, geometrically plausible, and interpretable set of unconditional crystal candidates and compress them into a structured representation suitable for further simulation and AI-guided prioritization.

3.1.2 Case 2: Symmetry-conditioned generation with the space_group model

To further assess conditional controllability in the MatterGen workflow of MatterMind, we performed a symmetry-conditioned generation experiment using the space_group model with a target space group of No. 225 (*Fm $\bar{3}$ m*), batch_size = 4, num_batches = 4, and diffusion_guidance_factor = 2.0. The aim of this case was not only to produce geometrically plausible crystal candidates, but also to examine whether the workflow could recover the requested symmetry target under strict post-processing evaluation while preserving structural diversity and downstream screenability. As in the previous case, MatterMind automatically executed the full generation-to-analysis pipeline, including structural ingestion, symmetry recognition, deduplication, descriptor extraction, rendering, and AI-assisted review.

From the perspective of workflow execution, the run was fully successful. All 16 generated structures were parsed without error, exported as CIF files, rendered as PNG images, and annotated with both SOAP summaries and Magpie features. No job-level warnings were recorded, and StructureMatcher deduplication again yielded 16 distinct groups, indicating that no duplicate or near-duplicate structures were identified within this small batch under the chosen matching tolerances. The full set of generated structures is shown in Fig. 5, and the corresponding screening summary is reported in Table 3. Thus, at the level of execution robustness, artifact completeness, and non-redundancy, the symmetry-conditioned workflow remained as stable as the unconditional baseline.

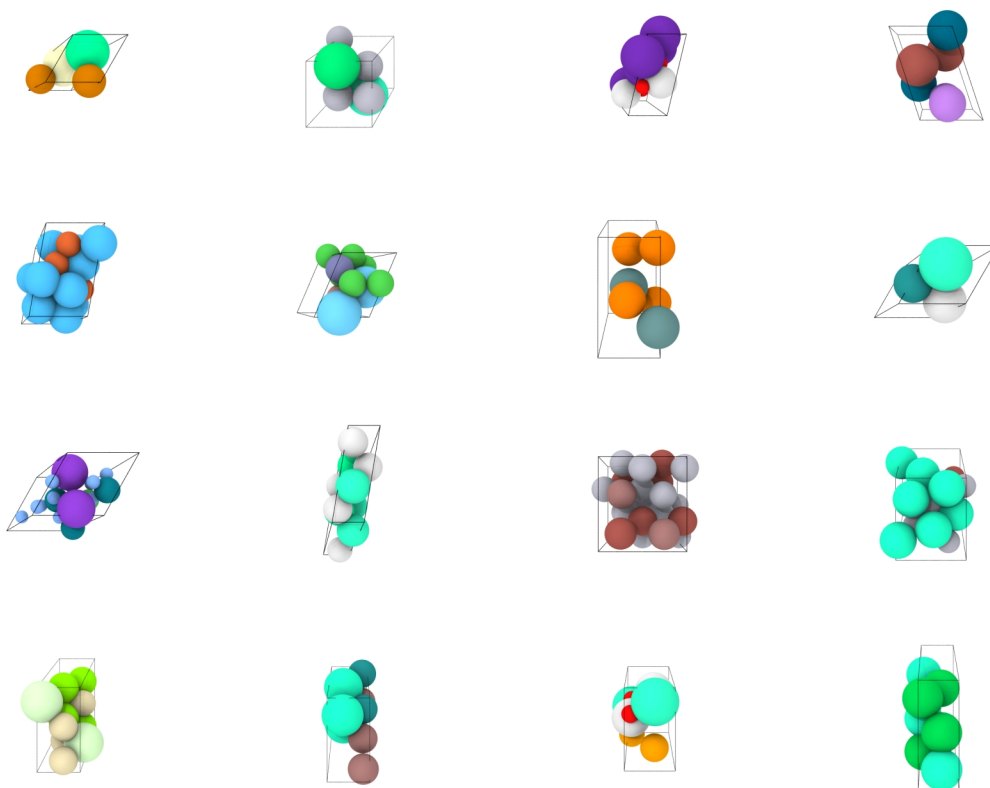


Figure 5. All 16 structures generated in the symmetry-conditioned space_group case and rendered by the MatterMind post-processing pipeline.

The run was conditioned on target space group No. 225 ($Fm\bar{3}m$) with batch_size = 4, num_batches = 4, and diffusion_guidance_factor = 2.0. Although all generated candidates were fully processed into inspectable artifacts and remained geometrically plausible under the platform screening criteria, none of the unrelaxed post-processed structures was assigned to the requested target space group under strict evaluation. The figure therefore illustrates both the structural diversity of the generated batch and the gap between nominal conditioning intent and realized symmetry outcomes in this experiment.

Table 3. Structured screening summary of the 16 structures generated in the symmetry-conditioned space_group case.

For each generated candidate, the table reports formula, atom count, post-processed symmetry assignment, volume per atom, minimum interatomic distance, close-contact statistics, deduplication-group identity, SOAP and Magpie feature coverage, and warning status. Together, these fields summarize how MatterMind converts raw symmetry-conditioned generative outputs into machine-readable evidence for geometry screening, symmetry analysis, diversity assessment, and downstream prioritization.

#	Formula	Atoms	Symmetry	V/atom	Min dist	Close contacts	Dedup	SOAP	Magpie	Warn
0	CeHoTe2	4	8 Cm	30.28	3.05 (Ho-Te)	0	G8 x1	92640	Yes	0
1	Tb2ErHg5	8	69 Fmmm	24.98	3.09 (Hg-Tb)	0	G10 x1	92640	Yes	0
2	Rb5Sc2O5	12	1 P1	27.41	1.95 (O-Sc)	0	G14 x1	92640	Yes	0
3	TlAsPd	6	1 P1	22.17	2.67 (As-Pd)	0	G2 x1	92640	Yes	0
4	Hf14Fe5	19	1 P1	19.44	2.30 (Fe-Fe)	0	G3 x1	92640	Yes	0
5	La2ZnGaNi6	10	1 P1	16.82	2.37 (Ni-Ni)	0	G12 x1	92640	Yes	0

#	Formula	Atoms	Symmetry	V/atom	Min dist	Close contacts	Dedup	SOAP	Magpie	Warn
6	GeP2	6	1 P1	23.19	2.27 (P-P)	0	G0 x1	92640	Yes	0
7	LiTbScRu	4	216 F-43m	19.53	2.94 (Ru-Tb)	0	G13 x1	92640	Yes	0
8	K2NiRh3F12	18	8 Cm	16.1	2.02 (F-Rh)	0	G15 x1	92640	Yes	0
9	Tb2HoSc3	12	8 Cm	28.39	3.26 (Ho-Sc)	0	G11 x1	92640	Yes	0
10	Tl3InHg6	20	1 P1	28.47	3.20 (Hg-In)	0	G1 x1	92640	Yes	0
11	Dy5(TlHg)2	18	2 P-1	28.52	3.01 (Hg-Hg)	0	G6 x1	92640	Yes	0
12	Pr3(MgPt)4	11	8 Cm	22.31	2.69 (Mg-Pt)	0	G7 x1	92640	Yes	0
13	Dy3In3Ru2	8	38 Amm2	22.84	2.89 (In-Ru)	0	G5 x1	92640	Yes	0
14	TbScSeO2	10	11 P2 ₁ /m	17.65	2.05 (O-Sc)	0	G9 x1	92640	Yes	0
15	DyTm2	9	189 P-62m	30.82	3.32 (Dy-Tm)	0	G4 x1	92640	Yes	0

The most important result of this case concerns conditioning fidelity. When the generated structures were evaluated using the post-processing symmetry assignments of the unrelaxed outputs recorded in metrics.json, none of the 16 candidates was assigned to the requested target space group No. 225 (Fm $\bar{3}$ m). This corresponds to a strict target-hit count of 0/16 and a hit rate of 0%. The batch nevertheless contained several geometrically sound and relatively high-symmetry structures, including one cubic $F\bar{4}3m$ candidate and one orthorhombic $Fmmm$ candidate, but these should be interpreted as attractive downstream screening candidates rather than as successful realizations of the specified Fm $\bar{3}$ m target. Accordingly, under the present sampling setting and strict post-processing criterion, the conditioning fidelity of this run was limited.

Geometric screening, however, showed a uniformly acceptable batch. All 16 structures passed the close-contact filter, with zero recorded close contacts throughout the run. The global minimum interatomic distance was 1.95 Å, and the median minimum distance was 2.79 Å, indicating that even the most compact candidates remained above the preset geometric sanity threshold. This is an informative result in itself, because it shows that symmetry conditioning did not lead to obviously pathological packing or short-range atomic overlap. In other words, although the requested target symmetry was not recovered under strict evaluation, the generated structures remained geometrically plausible and suitable for first-pass downstream screening.

Crystallographically, the generated batch exhibited a mixed symmetry distribution rather than a concentration in the requested target group. Specifically, six structures were assigned to $P1$ four to Cm and one each to $P\bar{1}$, $Amm2$, $P2_1/m$, $Fmmm$, $F\bar{4}3m$, and $P\bar{6}2m$. This pattern indicates that, in the present run, space-group conditioning acted more as a soft biasing mechanism than as a reliable hard constraint. At the same time, the appearance of several medium- and high-symmetry outputs remains useful from a workflow perspective, because these candidates can still be prioritized for follow-up relaxation and symmetry re-evaluation.

Among the generated structures, LiTbScRu (Index 7) is the most prominent follow-up candidate. It was assigned to the cubic space group $F\bar{4}3m$ and combines a small unit cell (four atoms), a moderate volume per atom of 19.53 Å³/atom, and a minimum interatomic distance of 2.94 Å. This combination of relatively high symmetry, compactness, and comfortable packing makes it especially attractive for subsequent first-principles validation. Tb₂ErHg₅ (Index 1) is another strong candidate, appearing in the orthorhombic space group $Fmmm$ with eight atoms per cell and similarly clean geometric indicators. Additional structures such as TbScSeO₂ (Index 14, $P2_1/m$) and Dy₃In₃Ru₂ (Index 13, $Amm2$) further illustrate that the batch contains several ordered and computationally tractable candidates even though strict target-space-group compliance was not achieved.

At the same time, the batch also contains a substantial low-symmetry component. The six $P1$ structures together with one $P\bar{1}$ structure account for nearly half of the generated set. These candidates are not geometrically invalid, but they are less immediately attractive for a first round of DFT validation because low symmetry in unrelaxed generative outputs may reflect either genuinely complex ordering or residual distortions that would only be resolved after structural relaxation. For example, $\text{Hf}_{14}\text{Fe}_5$ and $\text{La}_2\text{ZnGaNi}_6$ are both reasonably dense and geometrically acceptable, yet their triclinic $P1$ assignment makes their structural interpretation less straightforward than that of the cubic or orthorhombic candidates. This again highlights the practical value of MatterMind's post-processing layer: rather than treating all generated outputs as equivalent, the platform immediately distinguishes between symmetry-favorable candidates and those requiring more cautious interpretation.

A methodological limitation revealed by this case is that the user-specified target space-group number was not archived explicitly in the exported `metrics.json`, even though it was known from the run configuration. Consequently, the strict target-hit analysis reported here had to be reconstructed from the experimental setup and the post-processed symmetry labels, rather than being computed directly from job-level metadata. For future benchmark-oriented studies of symmetry-conditioned generation, the requested target space group should be stored explicitly in the structured output so that conditioning fidelity can be quantified automatically and reproducibly.

Overall, this case shows that the `space_group` workflow in MatterMind is effective as a symmetry-aware candidate generation and triage system, but that symmetry conditioning should be interpreted as a biasing mechanism rather than a guarantee of final crystallographic compliance. The batch was fully processable, non-redundant, and geometrically valid, and it yielded several attractive ordered candidates such as LiTbScRu and Tb_2ErHg_5 . However, the strict target-hit rate of 0% indicates that compliance with the requested $\text{Fm}\bar{3}m$ target was weak in this particular experiment. The most appropriate next step is therefore to subject the highest-symmetry and smallest-cell candidates to rapid relaxation and then re-evaluate their space-group assignments after optimization, so as to distinguish genuinely robust symmetry-conditioned outputs from structures whose apparent low symmetry is primarily a pre-relaxation artifact.

3.2 VASP Calculation

3.2.1 Case 1: Validation of the HDF5-Enabled VASP Workflow Using Cubic SrTiO_3

To validate the HDF5-enabled VASP workflow in MatterMind, cubic SrTiO_3 was selected as a representative validation case and studied through a three-step calculation sequence consisting of (i) structural relaxation, (ii) a static self-consistent calculation with density-of-states (DOS) analysis, and (iii) a line-mode band-structure calculation based on the relaxed structure and the charge density obtained from the static step. This material provides a suitable test bed because it combines a simple five-atom cubic perovskite structure with a well-known semiconducting electronic structure, allowing both numerical robustness and post-processing interpretability to be assessed in a controlled setting. When HDF5 support is enabled, VASP produces `vaspout.h5` as a hierarchical output file, and the VASP documentation explicitly recommends `py4vasp` for analyzing this file.

The relaxation step completed successfully and was automatically parsed by MatterMind into a structured `HDF5_metrics.json` record. The final structure retained the expected cubic perovskite symmetry, with space group $\text{Pm}\bar{3}m$ (No. 221), five atoms in the primitive cell, a lattice parameter of approximately 3.944 Å, and a cell volume of 61.34 Å³. The quality-control layer indicated clean termination, successful electronic and ionic convergence, and very small residual force and stress metrics, consistent with a stable relaxed configuration. The corresponding relaxation energy trace is shown in Fig. 6(a). These results demonstrate that the HDF5

workflow can transform a standard VASP relaxation into a machine-readable summary suitable for direct inspection and manuscript preparation, without manual parsing of OUTCAR or vasprun.xml.

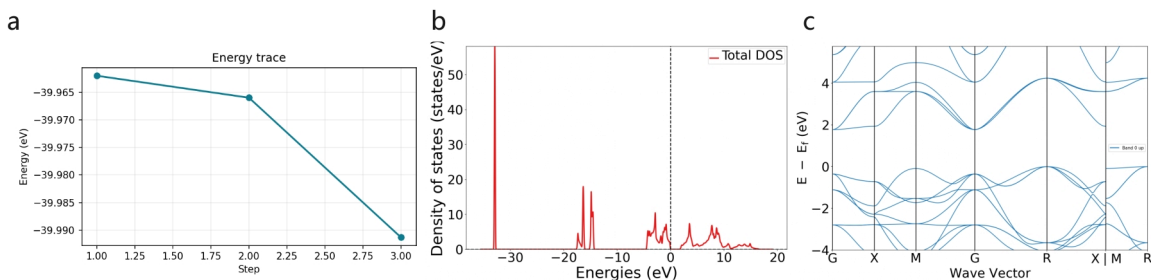


Figure 6. Validation of the HDF5-enabled VASP workflow in MatterMind using cubic SrTiO₃.

(a) Relaxation energy trace obtained during structural optimization, showing monotonic energy lowering toward the final relaxed configuration.

(b) Total density of states extracted from the static self-consistent calculation on the relaxed structure, with the Fermi level aligned at 0 eV.

(c) Line-mode band structure calculated from the relaxed structure using the charge density from the static step, showing an indirect band gap with the valence-band maximum near RRR and the conduction-band minimum near Γ along the selected high-symmetry path.

The static self-consistent calculation on the relaxed structure further confirmed the consistency of the workflow at the electronic-structure level. MatterMind again recovered the same cubic $Pm\bar{3}m$ symmetry and extracted the electronic descriptors required for downstream interpretation. In particular, the static post-processing plugin extracted an indirect band gap of 1.7734 eV from the electronic eigenvalue summary, together with Fermi-level information and DOS data. The corresponding DOS curve is shown in Fig. 6(b). Although this gap is substantially smaller than the experimental indirect gap of SrTiO₃ (3.25 eV; direct gap 3.75 eV), such underestimation is well known for semilocal DFT and should not be interpreted as a limitation of the MatterMind workflow itself. For the present study, the key result is that the platform extracted the band-gap information consistently and automatically from standard VASP outputs.

The band-structure step provided an additional test of workflow chaining and result interpretation. Starting from the relaxed structure and the charge density obtained in the previous steps, the line-mode band calculation yielded an indirect band gap of 1.7784 eV, in close agreement with the value extracted from the static step. Along the selected high-symmetry path, the valence-band maximum was located near RRR, whereas the conduction-band minimum was located near Γ as shown in Fig. 6(c). The close agreement between the static-step gap (1.7734 eV) and the band-step gap (1.7784 eV) indicates that the MatterMind HDF5 workflow provides internally consistent electronic-structure summaries across multiple stages of a standard VASP analysis sequence. The successful generation of band.png in the final step further demonstrates that the platform can recover not only scalar descriptors such as band gap and Fermi level, but also reusable visual outputs for scientific reporting.

An instructive aspect of this case is the behavior of the quality-control engine during the band calculation. The band step was labeled as degraded because the generic QC rule detected a large stress component and raised a STRESS_EXCEEDED warning. In this specific context, however, the warning should not be interpreted as a failure of the band-structure calculation itself. Line-mode band calculations are not intended to provide mechanically meaningful stress tensors, and the step still achieved clean termination and electronic convergence while producing a valid band plot and a consistent indirect gap. This observation highlights an important design implication for automated scientific workflows: QC flags should be interpreted in a workflow-specific manner, because warning types that are physically critical for relaxation tasks may be much less relevant for post-SCF electronic-structure evaluations.

Overall, the SrTiO₃ case demonstrates that the HDF5-enabled VASP workflow in MatterMind can support a realistic multistep first-principles pipeline from structural relaxation to DOS and band-structure analysis. Across the three stages, the platform preserved the structural identity of the system, extracted crystallographic, energetic, and electronic descriptors in a structured format, generated reusable plots, and exposed interpretable QC records for user review. Taken together, these results show that the HDF5 workflow in MatterMind functions not merely as a job-execution wrapper, but as a reproducible analysis layer that converts standard VASP calculations into structured, visual, and AI-ready scientific outputs.

3.2.2 Case 2: Validation of the VTST-Enabled VASP Workflow Using Oxygen-Vacancy Migration in 1×1×2 SrTiO₃

To further validate the VTST-enabled VASP workflow in MatterMind, an oxygen-vacancy migration problem in a 1×1×2 SrTiO₃ supercell was selected as a workflow-validation extension of the cubic SrTiO₃ HDF5 case presented in Section 3.2.1. The relaxed bulk SrTiO₃ structure obtained in the previous section was used as the starting point, after which two defective endpoint configurations were generated by removing one O atom and constructing a neighboring oxygen hop into the vacancy site. Because these script-generated endpoint structures were only initial guesses rather than fully relaxed local minima, this case specifically tested the endpoint-relax-first mode newly implemented in MatterMind for VTST calculations.

In this workflow, the user uploaded INCAR_endpoint, INCAR_neb, KPOINTS, POTCAR, POSCAR_i, and POSCAR_f, after which MatterMind automatically executed two endpoint relaxations in separate working directories, promoted the resulting CONTCAR files to relaxed NEB endpoints, generated the image path through VTST, launched the parent CI-NEB calculation, and finally collected the outputs into a structured post-processing record. The CI-NEB calculation was performed with three intermediate images (IMAGES = 3) and climbing-image optimization (LCLIMB = .TRUE.), using a 2×2×2 Gamma-centered k-point mesh and a 450 eV plane-wave cutoff. This case therefore validates not only the execution of VTST itself, but also the platform logic required to chain endpoint relaxation, image generation, parent-directory CI-NEB execution, and automated artifact parsing into a single reproducible workflow.

The resulting defective supercell corresponds to Sr₂Ti₂O₅ with nine atoms and lattice parameters inherited from the relaxed 1×1×2 SrTiO₃ cell, approximately 3.944 × 3.944 × 7.888 Å, corresponding to a cell volume of 122.68 Å³. Automatic structure analysis showed that the oxygen vacancy lowers the symmetry from the cubic bulk reference to *P4/mmm* for the relaxed initial endpoint and *Pmmm* for the relaxed final endpoint, while the highest-energy image along the path exhibits monoclinic *Pm* symmetry. This symmetry reduction is physically reasonable for a localized vacancy-migration process and shows that MatterMind can preserve and analyze structural identity not only for ideal bulk crystals, but also for low-symmetry defective pathway calculations.

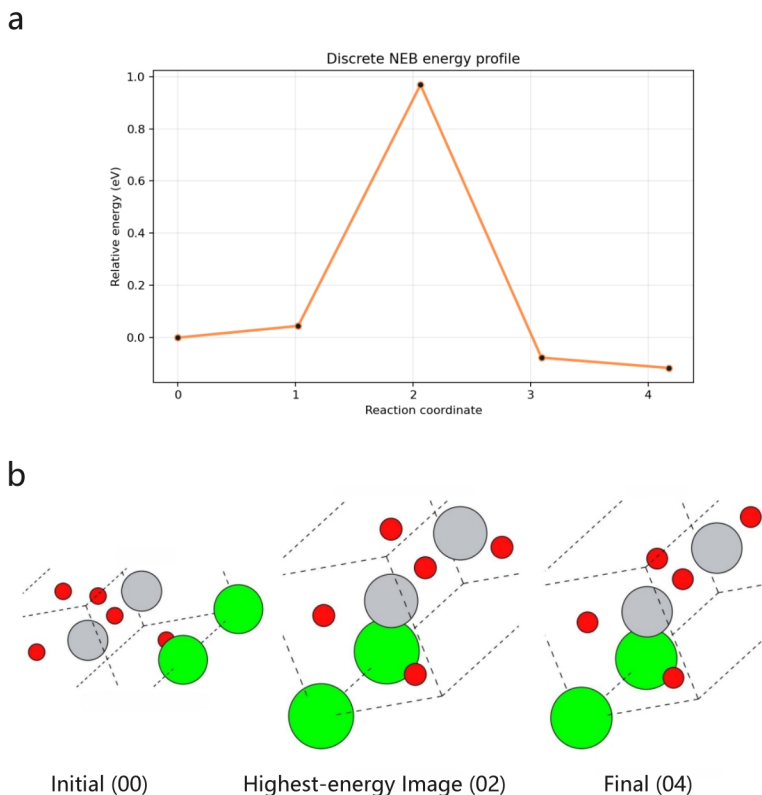


Figure 7. CI-NEB validation of the VTST-enabled VASP workflow in MatterMind using oxygen-vacancy migration in a $1 \times 1 \times 2$ SrTiO₃ supercell.

(a) Discrete CI-NEB energy profile obtained from the endpoint-relax-first workflow, showing a clear internal maximum at image 2 and a forward activation barrier of 0.9686 eV relative to the relaxed initial endpoint. The final endpoint lies 0.1163 eV lower in energy than the initial endpoint, corresponding to a reverse barrier of 1.0849 eV.

(b) Structural comparison among the relaxed initial endpoint, the highest-energy image (image 02), and the relaxed final endpoint. The highest-energy image is interpreted as a CI-NEB saddle-region candidate within the present path discretization and is consistent with the intended oxygen-vacancy hopping mechanism.

The discrete CI-NEB energy profile, shown in Fig. 7(a), exhibits a clear internal maximum at image 2, indicating that the workflow successfully bracketed a saddle-region candidate rather than producing a monotonic drift between endpoints. Relative to the relaxed initial endpoint, the highest-energy image lies 0.9686 eV higher, whereas the final endpoint is 0.1163 eV lower in energy than the initial one. Accordingly, the reverse barrier referenced to the lower-energy final state is 1.0849 eV. The path is therefore mildly exothermic overall but still contains a well-defined activated step. The atom associated with the largest endpoint displacement is an oxygen atom, with an endpoint-to-endpoint displacement of approximately 2.61 Å, consistent with the intended oxygen-vacancy hopping mechanism in the defective SrTiO₃ supercell. Because this calculation was performed in a compact $1 \times 1 \times 2$ defective supercell with a workflow-oriented parameter set, the reported barrier should be interpreted primarily as a validation result for the VTST-enabled pipeline rather than as a fully converged benchmark value for oxygen-vacancy diffusion in SrTiO₃.

A structural comparison among the relaxed initial endpoint, the highest-energy image, and the relaxed final endpoint is shown in Fig. 7(b). The highest-energy image should be interpreted as a CI-NEB saddle-region candidate within the present path discretization rather than as an independently verified transition state. Nevertheless, the resulting structural progression is chemically consistent with the intended vacancy-hopping event and provides an interpretable pathway representation for downstream analysis and reporting.

An instructive aspect of this case is the behavior of the automated QC and post-processing layer. MatterMind successfully generated the primary VTST artifacts, including the image-energy table, the raw NEB profile, and the endpoint-versus-highest-energy-image structural comparison. At the same time, the run was conservatively labeled as degraded because some compatibility-oriented image checks remained strict and because auxiliary interpolation-related outputs were not consistently available in this demonstration case. For workflow validation, however, the essential scientific outputs were recovered correctly: the endpoint-relax-first orchestration worked as intended, the CI-NEB path displayed an internal maximum, and the barrier profile was automatically converted into structured and visual outputs suitable for direct interpretation. This observation reinforces a broader design point already noted in Section 3.2.1, namely that QC records in automated scientific workflows should be interpreted in a workflow-specific manner, since conservative warnings do not necessarily invalidate the primary scientific result.

Overall, the SrTiO₃ oxygen-vacancy case demonstrates that the VTST-enabled VASP workflow in MatterMind can extend the platform from standard bulk HDF5 calculations to defect-migration and CI-NEB pathway analysis. Starting from a relaxed bulk structure, the platform supported scripted endpoint construction, automatic endpoint relaxation, CI-NEB path generation, structured extraction of barrier data, and generation of reusable plots for reporting. Taken together with the HDF5 case in Section 3.2.1, this result shows that MatterMind functions not merely as a job launcher, but as a workflow-aware analysis environment capable of organizing multistage first-principles calculations from equilibrium structures to activated diffusion pathways.

3.2.3 Case 3: Comprehensive Validation of the Wannier Workflow in Cubic SrTiO₃

3.2.3.1 Validation of the Wannier-Enabled VASP Workflow Through Orbital Localization and Compact Hamiltonian Extraction in Cubic SrTiO₃

To further extend the validation of MatterMind beyond the HDF5-based and VTST-based workflows, cubic SrTiO₃ was selected as a representative test system for the Wannier-enabled VASP workflow. Reusing the same material enables the Wannier workflow to be evaluated on the same structural and electronic reference system employed in the preceding HDF5 case, while extending the platform validation from direct first-principles outputs to localized-orbital and compact-Hamiltonian analysis. Cubic SrTiO₃ is well suited for this purpose because its near-gap electronic structure is dominated by chemically interpretable O 2p and Ti 3d states, making the resulting localization behavior straightforward to assess.

The workflow was organized into two chained stages, namely a static self-consistent calculation followed by Wannier post-processing. The SCF stage was performed for the five-atom cubic perovskite cell using a plane-wave cutoff of 520 eV and an 8×8×8 Gamma-centered k-point mesh, providing the charge density and wavefunctions required for the interface step. In the subsequent Wannier stage, MatterMind reused the generated CHGCAR and WAVECAR files, enabled the VASP-Wannier90 interface, and constructed a 14-orbital Wannier model (num_wann=14) from 32 Bloch bands (num_bands=32) using O:p and Ti:d initial projections. This case therefore validates not only the execution of Wannier90 itself, but also the platform logic required to chain SCF output reuse, interface preparation, orbital localization, Hamiltonian export, and automated post-analysis into a single reproducible workflow.

The Wannierization reached a converged final state with a total spread Ω_{total} of 11.5961 Å². For the 14 Wannier functions, this corresponds to an average spread of 0.8283 Å², while the maximum individual spread was 0.9727 Å², indicating a compact and well-localized basis. The localized orbital centers remained positioned near chemically expected O- and Ti-centered local environments, consistent with the intended O 2p and Ti 3d orbital manifolds. As shown in Fig. 8(a), the real-space distribution of Wannier centers is highly regular and remains tightly associated with the structural backbone of cubic SrTiO₃. The center offsets extracted by MatterMind are extremely small, with an average deviation on the order of 3.2×10^{-7} Å and a maximum deviation of approximately 3.6×10^{-7} Å from the nearest atomic backbone. These results indicate

that the workflow does not merely complete the Wannier calculation, but also recovers a chemically transparent localized-orbital description consistent with the bonding picture of cubic SrTiO₃.

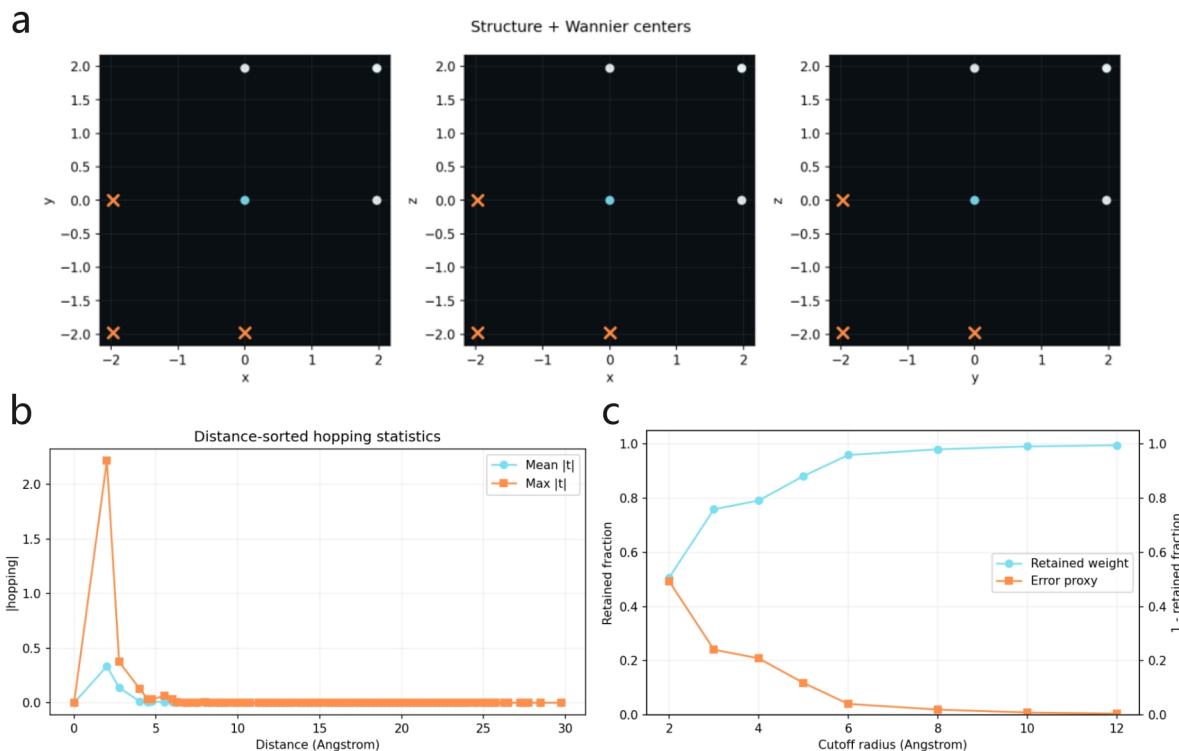


Figure 8. Wannier localization and compact Hamiltonian analysis for cubic SrTiO₃ in the Wannier-enabled MatterMind workflow.

(a) Overlay of the relaxed crystal structure and the extracted Wannier centers, showing that the localized orbital centers remain closely associated with chemically expected O- and Ti-centered local environments.

(b) Distance-sorted hopping statistics derived from the Wannier Hamiltonian, showing rapid decay of hopping magnitude with inter-center separation and a clear separation between dominant short-range terms and weaker long-range contributions.

(c) Hopping truncation analysis, expressed as retained hopping weight and its complementary error proxy as a function of real-space cutoff radius, demonstrating that the Hamiltonian is already highly compact at moderate truncation distances.

MatterMind further transformed the raw Wannier output into a structured tight-binding summary. The resulting Hamiltonian has a model dimension of 14 and is represented over 729 real-space lattice translations in the exported hr.dat data. The extracted hopping hierarchy is clearly nonuniform: the dominant nearest-neighbor hopping magnitude reaches 2.217 eV, whereas the next-nearest-neighbor scale is only 0.375 eV. As shown in Fig. 8(b), the distance-resolved hopping statistics display a strong short-range character, with the largest hopping amplitudes concentrated at small inter-center separations and rapidly diminishing with increasing distance. This trend is further quantified in Fig. 8(c), where the hopping truncation analysis shows that a real-space cutoff of 6 Å already retains about 95.96% of the total hopping weight, while a 10 Å cutoff retains about 99.13%. Together, these results indicate that the Wannierized Hamiltonian is not only localized in orbital space, but also compact in real space, making it suitable for downstream interpolation and model-based analysis.

An instructive aspect of this case is the interpretation of the quality-control record. The workflow was conservatively flagged as degraded, primarily because the disentanglement stage reached the preset maximum number of iterations and because the JSON Hamiltonian export was reduced to a summary representation for storage reasons. These warnings, however, do not invalidate the primary Wannierization result. The Wannier90 output explicitly reports a final state, the localization metrics remain compact, and MatterMind

assigned the run a high internal quality score of 96. As in the preceding validation cases, this example highlights the importance of workflow-aware QC interpretation: conservative warnings may remain visible while the primary scientific outputs remain valid and scientifically useful.

Overall, the cubic SrTiO_3 case demonstrates that the Wannier-enabled VASP workflow in MatterMind can support a realistic localized-orbital pipeline from self-consistent electronic-structure calculation to orbital localization and compact Hamiltonian analysis. The platform preserved the structural identity of the system, generated chemically meaningful Wannier centers, extracted quantitative localization and hopping descriptors, exported key Wannier artifacts, and produced reusable figures for scientific reporting. Taken together with the HDF5 and VTST validation cases, these results show that MatterMind functions not only as a job-execution interface, but also as a workflow-aware analysis environment capable of organizing diverse first-principles tasks into structured, visual, and AI-ready scientific outputs.

3.2.3.2 Validation of Wannier-Interpolated Band Reconstruction in Cubic SrTiO_3

To further validate the downstream usability of the Wannier-enabled VASP workflow in MatterMind, a Wannier-interpolated band-structure calculation was carried out for cubic SrTiO_3 using the 14-orbital Hamiltonian obtained in the previous subsection. The key validation question in this case is whether the compact real-space Hamiltonian derived from the O 2p- and Ti 3d-centered Wannier functions can faithfully reconstruct the electronic band dispersion along the standard high-symmetry path of the cubic Brillouin zone. Because the same cubic SrTiO_3 system was previously analyzed in the HDF5-based band-structure workflow in Section 3.2.1, this case also enables a direct comparison between the original DFT band description and the Wannier-interpolated one.

In this workflow, MatterMind reused the Wannier90 checkpoint and Hamiltonian files generated in the preceding Wannierization step and launched a post-Wannier band-interpolation task along the same high-symmetry path Γ -X-M- Γ -R-X-M-R. The calculation completed with status ok and without reported warnings, and the standard interpolation artifacts, including `wannier90_band.dat`, `wannier90_band.gnu`, and the corresponding band plot, were generated automatically. The interpolated path contains six line segments with dense sampling, yielding 757 k-points per band. This result therefore validates not only the execution of the interpolation step itself, but also the workflow-chaining logic required to proceed from self-consistent VASP output to Wannierization and then to Hamiltonian-based band reconstruction within a unified platform environment. The resulting interpolated band sequence is shown in Fig. 9.

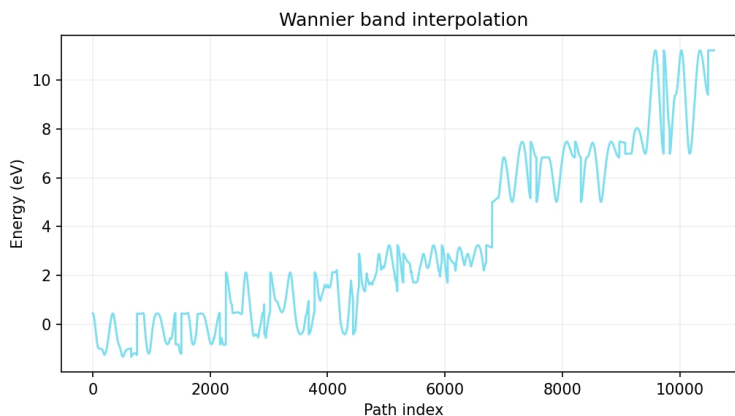


Figure 9. Wannier-interpolated band sequence for cubic SrTiO_3 generated from the 14-orbital real-space Hamiltonian in the MatterMind post-Wannier workflow.

The interpolated spectrum was evaluated along the cubic high-symmetry path Γ -X-M- Γ -R-X-M-R. The resulting band manifold reproduces the near-gap electronic structure of cubic SrTiO_3 , including the indirect R $\rightarrow$ Γ band gap discussed in the text, and serves as a downstream validation of the scientific usability of the Wannier-derived compact Hamiltonian.

Direct parsing of the interpolated band data shows that the resulting spectrum contains 14 bands, consistent with the 14-orbital Wannier model. The interpolated energies span approximately -1.33 to 11.23 eV in the plotting reference adopted by Wannier90. More importantly, within the physically relevant near-gap subspace spanned by the O 2p-dominated valence manifold and the Ti 3d-dominated conduction manifold, the interpolated band structure recovers an indirect band gap of 1.7783 eV. The valence-band maximum is located at R with an energy of 3.2369 eV, whereas the conduction-band minimum is located at Γ with an energy of 5.0152 eV. These band-edge locations and energies are fully consistent with the semilocal DFT reference obtained in Section 3.2.1.

A particularly important result is the quantitative agreement with the reference DFT band-structure analysis reported in Section 3.2.1. In that earlier HDF5-enabled case, MatterMind extracted an indirect gap of 1.7784 eV with the valence-band maximum at R and the conduction-band minimum at Γ . The present Wannier-interpolated result reproduces the same indirect $R \rightarrow \Gamma$ gap and the same band-edge locations, with a gap difference of less than 10^{-4} eV relative to the previous DFT summary. This near-identity demonstrates that the compact 14-orbital Hamiltonian is not only localized and computationally efficient, but also electronically faithful to the parent first-principles band structure along the relevant high-symmetry path.

This case also provides an instructive workflow-level implication. In the preceding Wannierization step, the run was conservatively flagged as degraded because the disentanglement stage reached the preset maximum number of iterations, even though the final Wannier functions were compact and chemically well positioned. The successful and quantitatively accurate band interpolation presented here shows that this warning did not prevent the resulting Hamiltonian from remaining scientifically useful for downstream analysis. More broadly, the SrTiO₃ band-interpolation case shows that the Wannier Hamiltonian generated within MatterMind remains a reliable basis for electronic-structure reconstruction, while the platform automatically produces reusable plots and numerical band descriptors for direct interpretation.

Taken together with the HDF5, VTST, and Wannier-localization validation cases, this result further supports the view that MatterMind functions as a workflow-aware scientific analysis layer rather than merely a calculation launcher. In the present case, the platform successfully carried the calculation chain from self-consistent DFT to Wannierization and finally to Hamiltonian-based band reconstruction, while preserving structured metadata, reusable artifacts, and task-specific post-analysis throughout the workflow.

3.2.3.3 Validation of Wannier-Interpolated DOS Reconstruction in Cubic SrTiO₃

To further validate the downstream usability of the Wannier-enabled workflow in MatterMind, a Wannier-based density-of-states (DOS) reconstruction was performed for cubic SrTiO₃ using the same 14-orbital Hamiltonian obtained in the previous subsection. This case complements the band-interpolation study by testing whether the localized real-space Hamiltonian can reproduce not only the path-resolved band dispersion, but also the reciprocal-space-integrated low-energy spectral distribution. Because the Wannier model was constructed from O p- and Ti d-centered orbital manifolds, the present calculation is intended to validate the near-gap electronic structure of cubic SrTiO₃, rather than the full all-electron DOS over a much wider energy range.

In this workflow, MatterMind reused the Wannier Hamiltonian and executed the DOS module using a $20 \times 20 \times 20$ interpolation mesh, an energy window from -8.0 to 8.0 eV, and an energy step of 0.02 eV. The task completed with status ok and without reported warnings, and the DOS data and plot were generated automatically. The resulting spectrum contains 801 sampled energy points, consistent with the prescribed energy window and step size. This case therefore validates the workflow logic required to proceed from self-consistent VASP output to Wannierization and then to Hamiltonian-based DOS reconstruction within a single chained platform environment. The interpolated DOS is shown in Fig. 10.

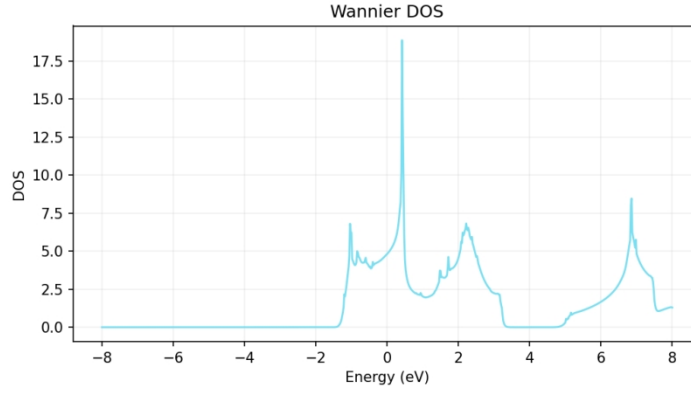


Figure 10. Wannier-reconstructed total density of states for cubic SrTiO₃ obtained from the 14-orbital Hamiltonian in the MatterMind post-Wannier workflow.

The DOS was evaluated on a $20 \times 20 \times 20$ interpolation mesh over the energy window from -8.0 to 8.0 eV with a step size of 0.02 eV. The spectrum displays a pronounced low-DOS region between the occupied and unoccupied manifolds, consistent with the semiconducting gap scale inferred from the Wannier-interpolated band structure discussed in the text.

The reconstructed DOS shows a well-defined low-density region between the occupied and unoccupied manifolds. In the native post-Wannier energy reference of the exported DOS file, the DOS is strongly suppressed between approximately 3.36 and 4.84 eV, and the value near the midpoint around 4.12 eV is essentially zero. Using an operational threshold of 0.5 states eV⁻¹, the low-DOS region extends from about 3.26 to 5.00 eV. This interval is in good agreement with the band-edge energies obtained from the Wannier-interpolated band-structure analysis in Section 3.2.4, where the valence-band maximum and conduction-band minimum were identified at 3.2369 and 5.0152 eV, respectively, corresponding to an indirect band gap of approximately 1.778 eV. The DOS reconstruction therefore provides an independent reciprocal-space-integrated confirmation of the near-gap depletion region and of the gap magnitude inferred from the interpolated band structure, although the DOS alone does not encode the momentum-space locations of the band edges.

The Wannier-derived DOS also reproduces the main low-energy spectral features of the SrTiO₃ reference calculation. In particular, the strongest valence-side DOS peak appears at approximately 0.42 eV in the post-Wannier energy reference, closely matching the prominent DOS peak at 0.4143 eV extracted from the HDF5-based static VASP analysis in Section 3.2.1. Additional occupied and unoccupied features are observed near -1.04 and 2.22 eV and near 5.16, 6.86, and 6.98 eV within the same reference. Because the present 14-orbital model consists of nine O p-like valence Wannier functions and five Ti d-like conduction Wannier functions, the occupied manifold can be interpreted as predominantly O-derived and the low-lying conduction manifold as predominantly Ti-d-derived. This assignment, however, should be understood as an inference from the Wannier basis construction and from the earlier DFT DOS analysis, rather than as a direct projected DOS decomposition in the present step.

Overall, the cubic SrTiO₃ DOS case demonstrates that the Wannier-derived Hamiltonian in MatterMind can reproduce not only the near-gap band topology along high-symmetry paths, but also the integrated low-energy spectral distribution relevant to semiconducting electronic-structure analysis. Together with the Wannierization and band-interpolation results, the present DOS reconstruction indicates that the 14-orbital SrTiO₃ model remains sufficiently faithful to recover the key low-energy spectral characteristics of the parent first-principles calculation in a structured and reusable form.

3.2.4 Case 4: Post-Wannier Quality Assessment for Ferromagnetic bcc Fe with SOC

To validate the post-SCF Wannier workflow implemented in MatterMind, we performed Wannier90 post-processing for ferromagnetic bcc Fe in the spinor formalism after the Wannier SCF step. The calculation

employed a primitive one-Fe bcc cell sampled on a Gamma-centered $12 \times 12 \times 12$ k-mesh. Initial projections were chosen as Fe s, p, and d orbitals, and 32 spinor Bloch states were disentangled into 18 Wannier functions. The purpose of this case was not yet to analyze response properties, but to assess whether the platform could construct a localized and reusable spinor Wannier representation suitable for subsequent postw90.x-based Berry-curvature, anomalous Hall conductivity, and Fermi-surface calculations. As summarized in Figure S4, Figure S5, and Figure S6, the run produced a physically interpretable Wannier model, although its real-space compactness remained only moderate rather than fully optimal.

The post-processing run completed successfully and generated the expected Wannier artifacts, including the Wannier output file, checkpoint, real-space Hamiltonian, center coordinates, hopping summaries, and structured visualization products. Wannierisation formally converged, with final spread components of $\Omega_I = 10.3364 \text{ \AA}^2$, $\Omega_D = 0.0107 \text{ \AA}^2$, $\Omega_{OD} = 6.0257 \text{ \AA}^2$, and $\Omega_{\text{total}} = 16.3728 \text{ \AA}^2$. The average spread was $0.9113 \text{ \AA}^2$, while the maximum spread reached $1.9982 \text{ \AA}^2$. As shown in Figure 11a, the spread distribution indicates that the resulting Wannier basis is overall reasonably localized, although a subset of functions remains noticeably more extended than the most compact orbitals in the set.

The real-space center analysis supports the physical interpretability of the constructed basis. All Wannier centers were assigned to Fe as the nearest atomic site, with an average center offset of $1.97 \times 10^{-4} \text{ \AA}$ and a maximum offset of $1.08 \times 10^{-2} \text{ \AA}$. Thus, the centers remain spatially concentrated around the Fe site rather than drifting into clearly interstitial regions. For a SOC-enabled spinor calculation, this is an encouraging sign that the chosen projection manifold retains an atomic-like interpretation despite band entanglement and spin-orbit mixing. This behavior is visualized in Figure 11b, where the Wannier centers cluster closely around the underlying Fe framework in the projected cell views.

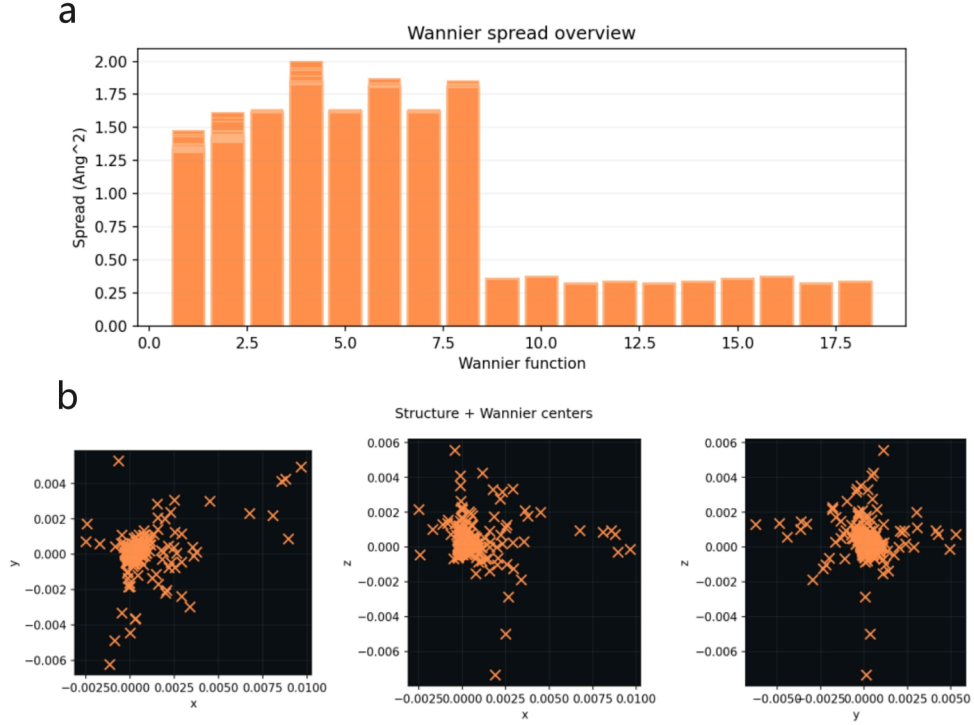


Figure 11. Localization and Wannier-center diagnostics for ferromagnetic bcc Fe with SOC.

(a) Spread overview of the 18 spinor Wannier functions obtained from the Fe s, p, d projection manifold after disentanglement of 32 spinor Bloch states.

(b) Projected views of the bcc Fe primitive cell overlaid with Wannier centers, showing that the centers remain clustered around the Fe site rather than drifting into strongly interstitial regions.

At the same time, the model is not fully ideal and should be interpreted with appropriate caution. The output reported that the maximum number of disentanglement iterations was reached, and the structured post-analysis marked the result as degraded despite formal convergence. This qualification is consistent with the relatively large off-diagonal spread Ω_{OD} and with the diffuse real-space Hamiltonian. The tight-binding model contains 1909 lattice-translation points and 618516 hopping terms in total, indicating that the resulting Wannier Hamiltonian is usable but not especially compact in a strict truncation sense.

The hopping diagnostics make this limitation more concrete. The nearest-neighbor shell at 2.5 Å has a mean absolute hopping magnitude of 0.233 eV and a maximum of 1.714 eV, while the next-nearest-neighbor shell at 2.75 Å still reaches a maximum of 2.196 eV. Moreover, truncating the Hamiltonian at 6 Å retains only 88.24% of the total absolute hopping weight, whereas a 10 Å cutoff is required to retain 94.30%. As shown in Figure 12a-b, the dominant hopping amplitudes are concentrated at relatively short range, but longer-range contributions remain non-negligible and should not be ignored in precision-sensitive interpolation tasks.

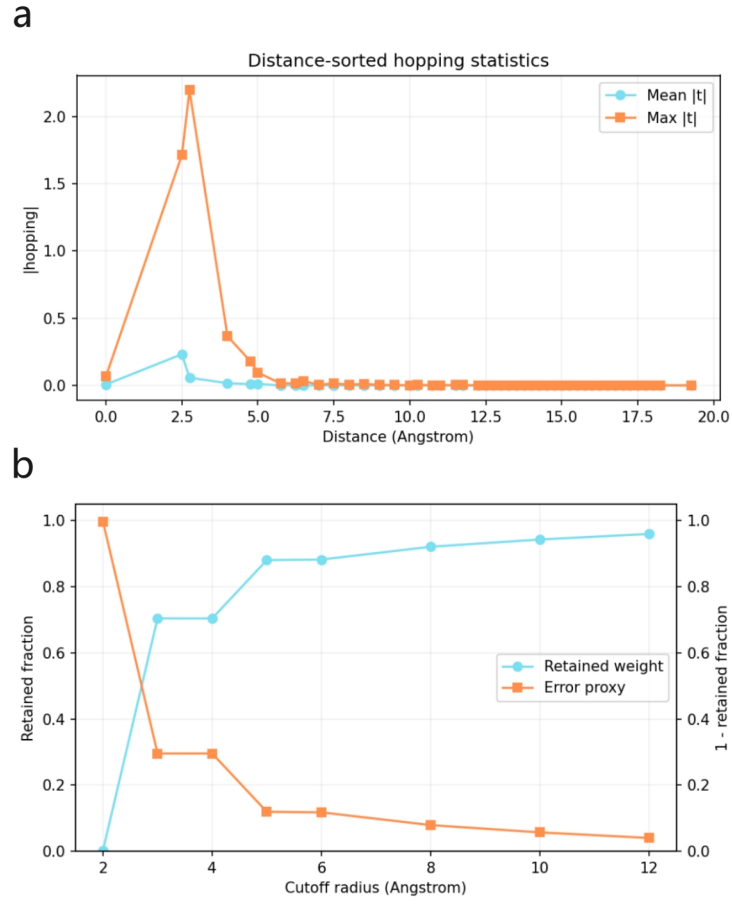


Figure 12. Real-space compactness diagnostics of the spinor Wannier Hamiltonian for ferromagnetic bcc Fe with SOC. (a) Distance-sorted hopping statistics, showing the mean and maximum absolute hopping magnitude as a function of real-space distance. (b) Retained hopping weight and corresponding truncation error proxy as functions of cutoff radius, illustrating that the Hamiltonian is usable but not highly compact in a strict truncation sense.

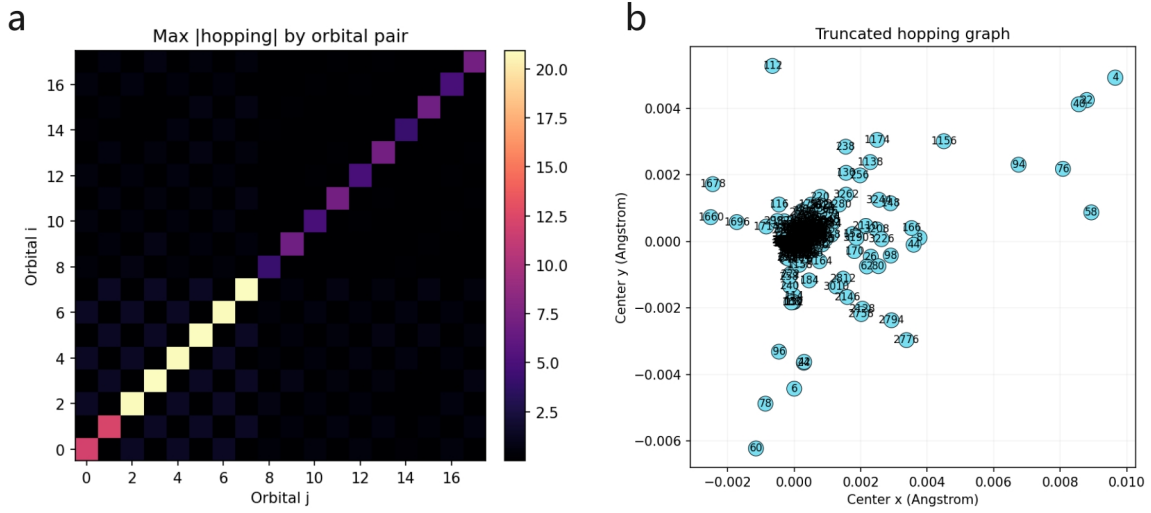


Figure 13. Secondary orbital-coupling diagnostics for the spinor Wannier Hamiltonian of ferromagnetic bcc Fe with SOC.

(a) Pair-resolved heat map of the maximum absolute hopping magnitude between Wannier orbital pairs.

(b) Truncated hopping graph representation of dominant real-space hopping channels, showing a dense central cluster together with a smaller number of more extended couplings.

A more detailed view of the orbital-coupling structure is provided in Figure 13. The pair-resolved hopping heat map (Figure 13a) shows that the strongest couplings are not uniformly distributed across the 18-function basis, while the truncated hopping graph (Figure 13b) highlights a dense central cluster together with a smaller number of more extended channels. These outputs are most naturally interpreted as secondary diagnostics rather than as primary performance indicators, but they are useful for confirming that the Hamiltonian contains both dominant local channels and a non-negligible tail of longer-range couplings.

An additional practical point concerns visualization. Because the SOC workflow is associated with LSORBIT-enabled handling, the platform did not rely on volumetric Wannier-orbital cube plots in the final analysis workflow. Instead, the post-Wannier assessment focused on center overlays, spread statistics, and hopping diagnostics. Even without real-space isosurface plots, these outputs are sufficient to diagnose localization quality, spatial center placement, orbital coupling patterns, and truncation behavior. In this sense, the present case provides the essential bridge between the SCF-generated Bloch manifold and the later postw90.x calculations.

Taken together, this case demonstrates that the MatterMind Wannier workflow can successfully produce a converged and reusable spinor Wannier representation for ferromagnetic bcc Fe with SOC. The resulting 18-function model is physically interpretable and operationally suitable for downstream interpolation-based analyses, but it is not fully optimal in compactness: the disentanglement warning, the sizeable Ω_{OD} , and the persistence of longer-range hopping terms all indicate that the model should be regarded as scientifically usable rather than maximally localized. This level of quality is nevertheless sufficient for the next-stage Berry/AHC and Fermi-surface calculations discussed below.

3.2.5 Case 5: Validation of Advanced Post-Wannier Analyses in Ferromagnetic bcc Fe with SOC

3.2.5.1 Berry-Curvature-Derived Anomalous Hall Conductivity Analysis for Ferromagnetic bcc Fe with SOC

To evaluate whether the SOC-enabled Wannier Hamiltonian obtained in Case 1 is sufficiently robust for interpolation-based response calculations, we next performed a Berry-curvature-derived anomalous Hall

conductivity (AHC) calculation with postw90.x for ferromagnetic bcc Fe. The calculation was carried out on the same 18-Wannier-function spinor model, using a $60 \times 60 \times 60$ Berry-interpolation mesh and a chemical-potential scan from -0.5 to 0.5 eV in steps of 0.02 eV, yielding 51 sampled energy points and three reported Cartesian AHC components.

The run completed successfully with no recorded warnings in the structured metrics, and the workflow generated the expected post-processing outputs, including the Berry/AHC summary, the detailed wpout report, and the AHC-versus-chemical-potential figure. In the postw90.x output, no k points triggered adaptive refinement at any scanned energy. At the workflow level, this confirms that MatterMind can take the SOC-enabled Wannier model produced in the previous step and execute an energy-resolved AHC calculation without manual file editing.

The resulting AHC response is clearly dominated by a single component. As shown in Figure 14, the z component is the largest throughout the scanned window. It increases from 0.0376 S/cm at -0.5 eV to 0.3205 S/cm at the reference Fermi level and then rises steadily on the positive-energy side, reaching 1.9790 S/cm at $+0.5$ eV, which is the largest absolute AHC value recorded in the scan. By contrast, the x component remains negative and comparatively small, varying only between about -0.1174 and -0.0387 S/cm, while the y component stays positive and modest, ranging from 0.0299 to 0.1034 S/cm. The main qualitative outcome is therefore not a uniformly strong Hall response in all directions, but a pronounced enhancement of one dominant channel under positive chemical-potential shift.

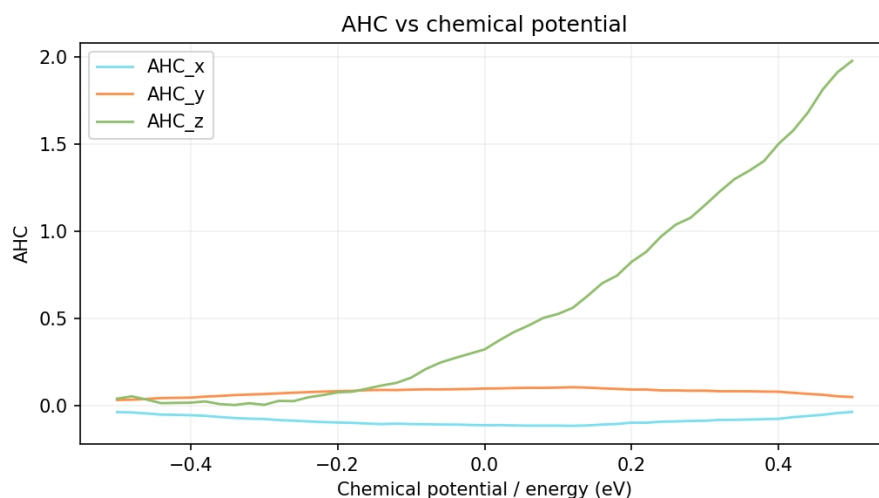


Figure 14. Berry-curvature-derived anomalous Hall conductivity of ferromagnetic bcc Fe with spin-orbit coupling, computed with postw90.x from the 18-function spinor Wannier Hamiltonian. The three reported Cartesian components are plotted as functions of chemical potential over the range from -0.5 to 0.5 eV using a $60 \times 60 \times 60$ interpolation mesh. The z component dominates the response and increases strongly for positive chemical-potential shifts, reaching the largest magnitude at the upper end of the scan window, whereas the x and y components remain comparatively small. The figure shows that the MatterMind workflow can produce an energy-resolved AHC trend directly from the Wannierized electronic structure while also highlighting the need for further convergence testing before quantitative interpretation.

This behavior is scientifically useful in two respects. First, it shows that the Wannier representation is operationally capable of resolving a continuous and nontrivial AHC trend over a finite energy window, rather than producing obviously irregular interpolation artifacts. Second, it suggests that the anomalous Hall response of this Wannierized bcc Fe model is sensitive to band filling in an asymmetric manner: hole-side shifts leave all components relatively small, whereas electron-side shifts progressively amplify the dominant component. Within the rigid-band picture adopted here, the positive-energy side is therefore the most responsive region of the present scan window and the most natural target for follow-up verification.

At the same time, this case should be interpreted as a trend-level interpolation result rather than a fully converged transport prediction. Only a single Berry-interpolation mesh, $60 \times 60 \times 60$, was tested in the present run, and no convergence study with respect to k-mesh density, smearing settings, or Wannier-model variation has yet been performed. Accordingly, the present data are sufficient to establish successful workflow execution and to identify the overall shape and relative dominance of the AHC components, but not to claim a fully converged anomalous Hall conductivity for ferromagnetic bcc Fe. In particular, the strong rise of the dominant component near the upper end of the scan window should be regarded as a physically suggestive trend that merits verification on denser meshes before being used for quantitative comparison with literature or experiment.

Taken together, this case demonstrates that the MatterMind Wannier and postw90.x workflow can successfully bridge from an SOC-enabled Wannier Hamiltonian to an energy-resolved AHC analysis for ferromagnetic bcc Fe. The calculation produced a continuous three-component AHC scan over 51 chemical-potential points, revealed a clear dominant response channel, and identified a strong enhancement on the positive-energy side. This is sufficient to justify proceeding to the next post-Wannier analysis stage, while also making clear that convergence testing remains necessary before stronger quantitative claims are made.

3.2.5.2 Wannier-Interpolated Fermi-Surface Export and XCrySDen Visualization for Ferromagnetic bcc Fe with SOC

To complete the post-Wannier validation workflow, we next examined whether the SOC-enabled Wannier Hamiltonian obtained for ferromagnetic bcc Fe could support direct Fermi-surface export and external visualization. Starting from the same 18-function spinor Wannier model used in Cases 1 and 2, we ran the Fermi-surface post-processing step with `restart = plot`, `fermi_surface_plot = true`, `fermi_surface_num_points = 120`, and `fermi_surface_plot_format = xcrysden`, using a Fermi energy of 5.4986400355 eV. The run completed successfully with status ok and no recorded warnings, and the workflow generated the expected `wannier90.bxsf` output file. Unlike the AHC case, the present module is validated primarily through successful BXSf generation and reciprocal-space visualization rather than through a dense set of scalar post-processing metrics.

Using the exported BXSf file, we visualized the Wannier-interpolated Fermi surface in XCrySDen²⁵. At the selected Fermi energy, the XCrySDen band selection identified bands 5 through 10 as the bands contributing to the plotted Fermi surface. As shown in Figure 15a, the merged visualization reveals a clearly multiband Fermi surface within the displayed Brillouin-zone frame, with the total topology composed of both extended sheet-like features and compact closed pockets. This is an important practical result because it shows that the interpolated Hamiltonian is not only sufficient for scalar response calculations such as AHC, but also capable of supporting direct reciprocal-space visualization of the metallic states.

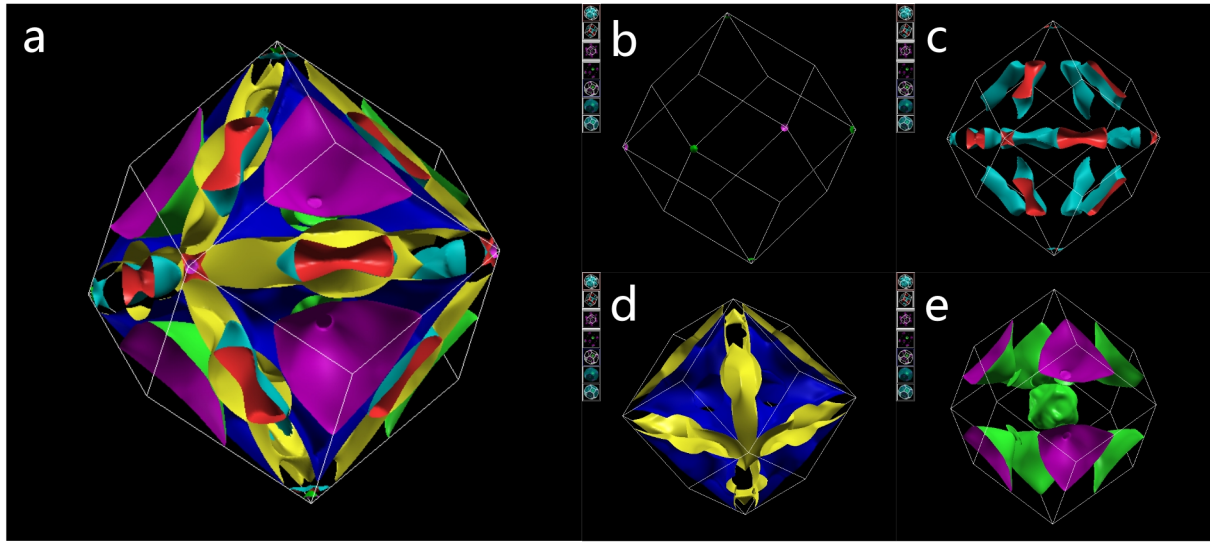


Figure 15. XCRYSDen visualization of the Wannier-interpolated Fermi surface of ferromagnetic bcc Fe with spin-orbit coupling. The Fermi surface was exported by MatterMind as a wannier90.bxsf file using a Fermi energy of 5.49864 eV and then visualized externally in XCRYSDen. Bands 5 through 10 were identified as the crossing bands at the selected Fermi energy; the merged view and representative band-resolved panels together show that the Fermi surface is multiband in character and contains both extended sheet-like features and compact pocket-like contributions. This figure demonstrates that the MatterMind Wannier workflow can produce a visualization-ready Fermi-surface representation directly from the SOC-enabled Wannier Hamiltonian.

- (a) Merged Fermi-surface view including all selected crossing bands, showing the overall multiband topology within the displayed Brillouin-zone frame.
- (b) Band 5 contribution, consisting mainly of small isolated pockets.
- (c) Band 6 contribution, showing more extended necked surfaces along symmetry-related directions.
- (d) Band 7 contribution, forming broad sheet-like features over a large portion of the displayed zone.
- (e) Band 8 contribution, combining a compact central pocket with larger surrounding surface segments.

The representative band-resolved XCRYSDen panels further show that the crossing bands contribute in qualitatively different ways. Figure 15b shows that band 5 contributes only a few small isolated pockets. Figure 15c shows that band 6 forms more extended, necked surfaces distributed along symmetry-related directions. Figure 15d shows that band 7 generates broad sheet-like features spanning large regions of the displayed zone. Figure 15e shows that band 8 combines a compact central pocket with larger surrounding surface segments. Taken together with the merged view, these panels indicate that the total Fermi surface is not dominated by a single simple sheet, but instead results from the superposition of several geometrically distinct band contributions. The remaining crossing bands, bands 9 and 10, contribute comparatively compact pocket-like features and are therefore not shown separately in the representative figure set.

From the perspective of methodological validation, this case is significant because it confirms that the Wannier Hamiltonian remains usable beyond post-processing diagnostics and Berry-curvature-based response scans. The successful BXSf export and the physically nontrivial XCRYSDen surfaces together provide an additional qualitative check on the interpolation pipeline. In practical terms, they show that the platform can deliver a visualization-ready reciprocal-space representation that the user can directly inspect for band-resolved metallic features without having to reconstruct the export workflow manually.

At the same time, the present case should be interpreted as a qualitative visualization demonstration rather than a complete Fermiology study. The current outputs establish that a usable Wannier-interpolated Fermi surface was generated and that bands 5 through 10 contribute at the chosen Fermi level, but they do not by themselves establish detailed pocket classification, carrier character, nesting behavior, or quantitative agreement with direct DFT Fermi-surface calculations or experiment. Such claims would require denser validation, explicit

band-by-band comparison to the underlying electronic structure, and possibly additional convergence checks with respect to the Fermi-surface sampling density. Within the scope of the present work, the main conclusion is therefore that MatterMind can robustly export and support visualization of a nontrivial multiband Fermi surface for ferromagnetic bcc Fe with SOC.

4 Conclusions

In this work, we developed MatterMind, a web-based workflow orchestration platform that integrates crystal generation, first-principles simulation, automated post-processing, visualization, and LLM-assisted interpretation into a unified environment for materials research. Rather than serving as a simple web wrapper for external codes, MatterMind was designed as a workflow-aware scientific analysis platform in which heterogeneous computational tasks are executed in dedicated runtime environments, transformed into structured metrics, and returned as directly inspectable research outputs. By combining a React/FastAPI/Celery/Redis software stack with domain-specific pipelines for MatterGen, VASP, VTST, Wannier90, and postw90.x, the platform provides a consistent interface for task submission, live log streaming, artifact management, structured quality control, and AI-guided interpretation.

The results demonstrate that MatterMind can support multiple representative materials-science workflows in a reproducible and interpretable manner. In the MatterGen branch, unconditional and conditional generation experiments showed that raw outputs can be converted into nonredundant, geometrically screened, symmetry-annotated, and descriptor-enriched candidate sets suitable for downstream triage. In the HDF5-enabled VASP workflow, the cubic SrTiO₃ case confirmed that structural, energetic, and electronic information can be extracted automatically from standard DFT calculations and transformed into reusable metrics and publication-oriented plots. In the VTST workflow, the oxygen-vacancy migration example demonstrated that MatterMind can orchestrate endpoint relaxation, CI-NEB path construction, barrier extraction, and workflow-aware QC within a single automated pipeline.

The Wannier-enabled branch further showed that MatterMind can support both baseline interpolation validation and advanced post-Wannier analyses. In cubic SrTiO₃, the platform generated localized Wannier functions, constructed a compact real-space Hamiltonian, and reproduced band-interpolated and DOS-derived electronic features with high fidelity to the parent first-principles calculation. In ferromagnetic bcc Fe with spin-orbit coupling, MatterMind produced a reusable spinor Wannier representation, diagnosed its localization quality and Hamiltonian compactness, and extended the same model to Berry-curvature-derived anomalous Hall conductivity analysis and Wannier-interpolated Fermi-surface export and visualization. These results show that MatterMind can organize diverse computational tasks into structured, visual, and AI-ready scientific outputs rather than isolated code executions.

A central implication of this work is that structured post-processing and workflow-specific QC are essential for trustworthy automation in computational materials science. Across the HDF5, VTST, Wannier, and post-Wannier cases, some warnings proved conservative rather than scientifically fatal, whereas in generative workflows apparently successful runs could still miss the intended physical condition. MatterMind addresses this gap by making such discrepancies explicit through metrics-driven interpretation rather than leaving users to infer validity from raw logs or native files alone.

At the same time, this study should be viewed primarily as a platform-validation study rather than an attempt to establish definitive benchmark values for every physical problem considered. Some demonstration cases used compact cells, moderate interpolation settings, or workflow-oriented parameters chosen to validate module chaining and downstream usability. Future work will therefore focus on systematic convergence studies, broader benchmarking against literature-quality reference calculations, and tighter integration of Wannier/postw90.x response-property analysis. Additional development will also target literature-aware AI

benchmarking and web-integrated reference comparison, so that computed results can be more directly contrasted with trusted published or database reference values. On the usability side, we plan to extend the AI layer toward interactive pre-run guidance for uploaded VASP input files, as well as to improve front-end local model loading and unified model management for both local and cloud-based models. In the generative branch, future work will further extend the platform from the current generation + screening paradigm toward more closed-loop validation and candidate evaluation, including tighter coupling between generation, post-generation filtering, and downstream stability assessment. The LLM-assisted interpretation layer can also be further strengthened through uncertainty-aware prompting and deeper coupling between structured metrics and automated scientific reporting.

Overall, MatterMind demonstrates that a unified platform can bridge materials generation, first-principles simulation, transition-state analysis, localized-orbital modeling, response-property interpolation, reciprocal-space visualization, and AI-assisted interpretation within one coherent workflow environment. By narrowing the gap between code execution and scientific interpretation, the platform provides a practical foundation for more automated, transparent, and extensible materials-discovery pipelines.

ACKNOWLEDGEMENTS

This work was supported by the Jiangsu Provincial Science and Technology Project Basic Research Program (Natural Science Foundation of Jiangsu Province) (No. BK20211283). The authors are also indebted to the National Supercomputer Center in Guangzhou for providing high-performance computational resources and the staff at the ESRF Grenoble for the assistance and support in using the BM23 beamline (proposal No.CH7455).

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Author Contributions

The author has given approval to the final version of the manuscript.

Data Availability Statement

The data that supports the findings of this study are available within the article.

Conflict of Interest

The authors have no conflicts to disclose.

Data and Software Availability Statement

The MatterMind platform developed in this work is available as an open-source project on GitHub at <https://github.com/Yujun-Lu/MatterMind>. Due to licensing restrictions and the presence of sensitive information in parts of the workflow artifacts, the full experimental files have not been uploaded to the public repository. However, the authors are willing to provide the experimental data, together with the relevant input and output files needed for reproduction of the results reported in this study, upon reasonable request. For

readers or reviewers who wish to directly experience the current prototype, SSH-based access and basic usage guidance can also be provided upon request.

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